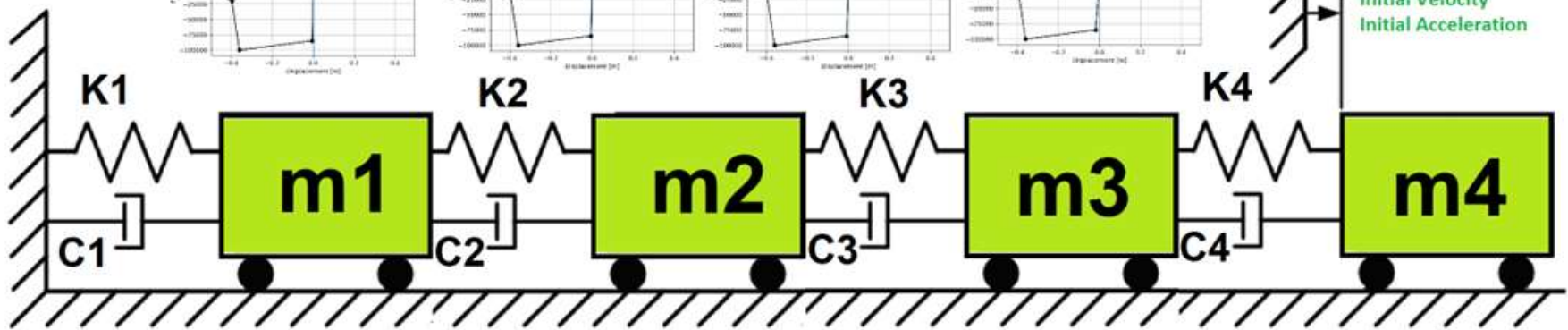
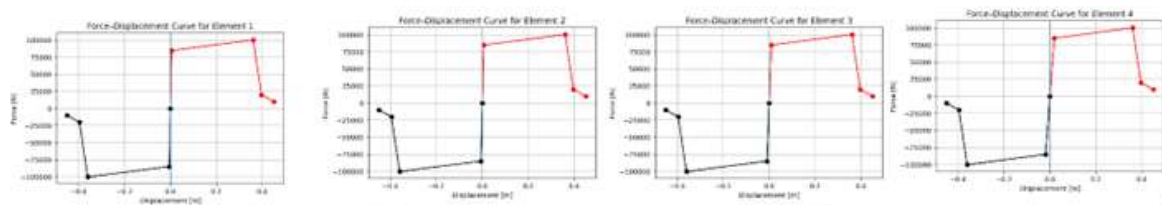


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

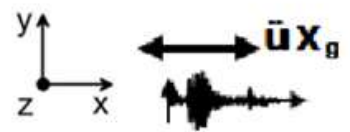
NONLINEAR DYNAMIC ANALYSIS UNDER FREE-VIBRATION COMPUTATION AND VISUALIZATION RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING OPENSEES

(CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM)

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



Initial Displacement
Initial Velocity
Initial Acceleration

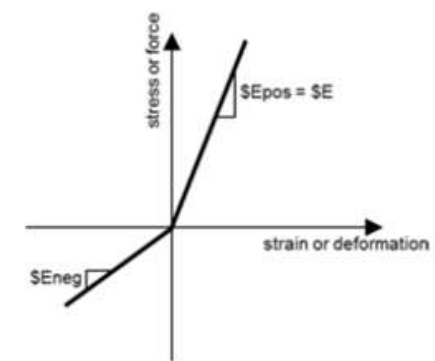
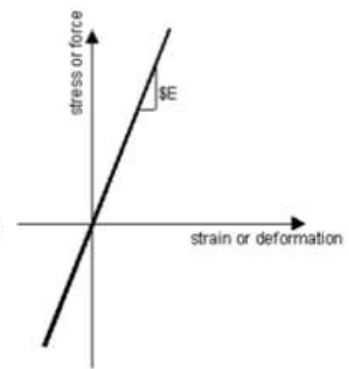


$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

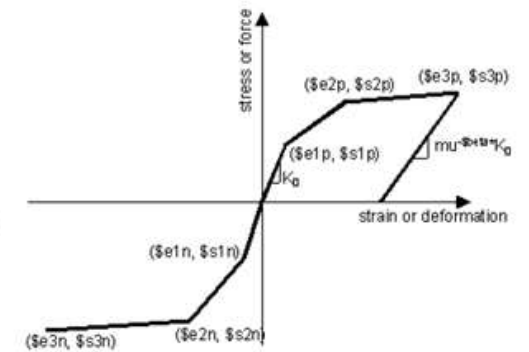
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

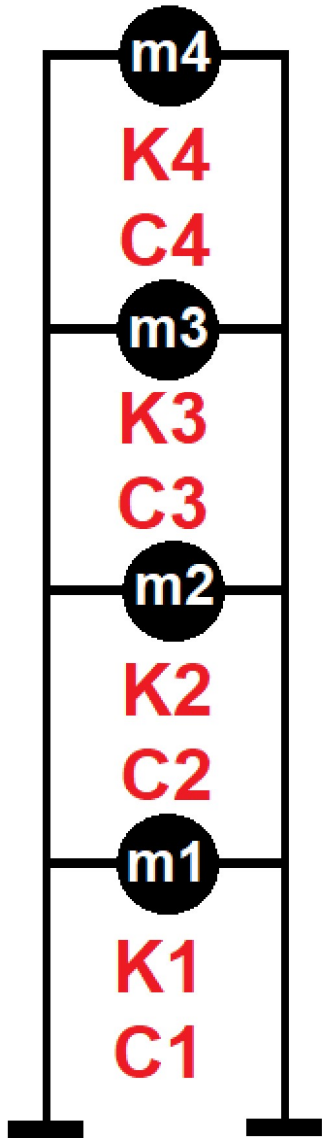
Δ_u = Lateral Ultimate Displaement from Pushover Analysis



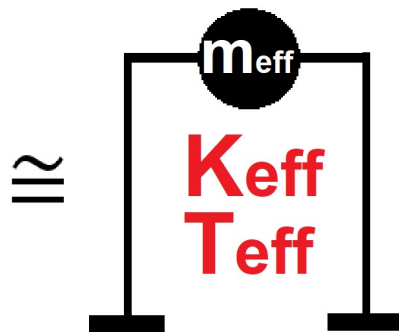
ELASTIC MATERIAL



INELASTIC MATERIAL

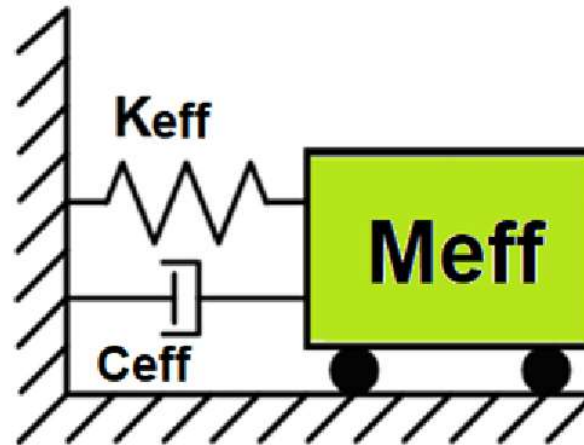


Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

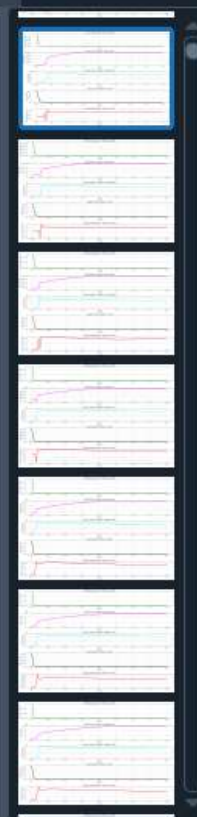
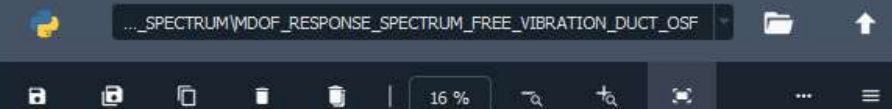
C:\Users\Dell\Desktop\OPENSEES_FILES\CONSTANT_STR...MDOF_RESPONSE_SPECTRUM_FREE_VIBRATION_DUCT_OSF.py

MDOF_RESPONSE_SPEC...RATION_DUCT_OSF.py X

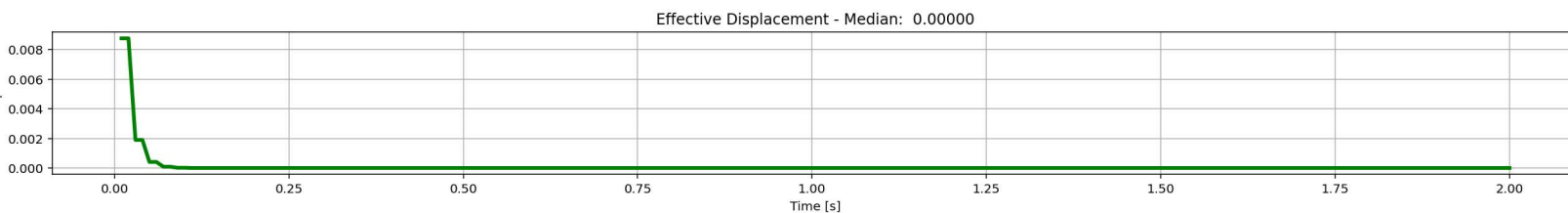
```

1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # NONLINEAR DYNAMIC ANALYSIS UNDER FREE-VIBRATION AND VISUALIZATION
4 # RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING C
5 #
6 # INVESTIGATION OF THE EFFECTS OF MDOF STRUCTURAL DUCTILITY ON DAMAGE LEVEL AND STRUCTURAL
7 # USING NONLINEAR DYNAMIC ANALYSIS
8 #
9 #
10 #-----
11 # CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM
12 #-----
13 # EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH FREE
14 #
15 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
16 #-----
17 # IT IS RECOMMENDED FOR BETTER SEEING THE STRUCTURAL DUCTILITY AND STRUCTURAL OVER-STRENGTH F
18 # INITIAL DISPLACEMENT HAVE TO GREATER THAN YIELD DISPLACEMENT.
19 #-----
20 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
21 # EMAIL: salar.d.ghashghaei@gmail.com
22 #####
23 """
24 Investigation of the Effects of Structural Ductility on Damage Level and Structural Damping:
25 This code implements a comprehensive nonlinear dynamic analysis framework for
26 performance-based earthquake engineering assessment of single-degree-of-freedom
27 (SDOF) systems. The methodology combines traditional nonlinear time-history
28 analysis with modern probabilistic and machine learning techniques for advanced
29 structural performance evaluation with changing structural ductility ratio and over strength f
30
31 KEY ENGINEERING OBJECTIVES:
32 1. Comparative assessment of hysteretic models for free-vibration response prediction
33 2. Probabilistic free-vibration demand analysis using multiple ground motions
34 3. Development of fragility curves for performance-based earthquake engineering

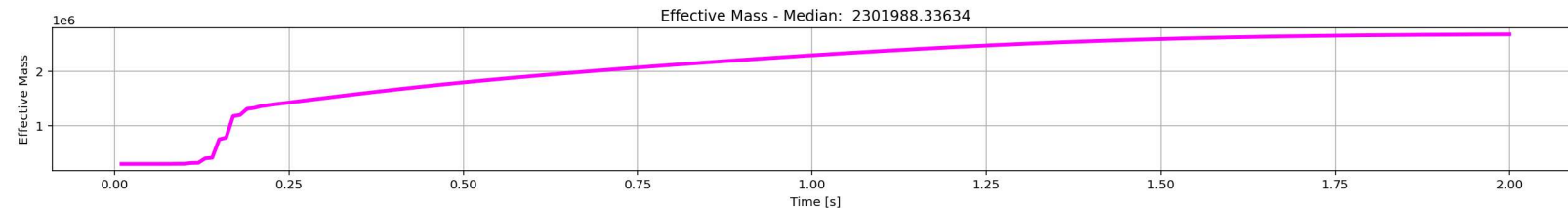
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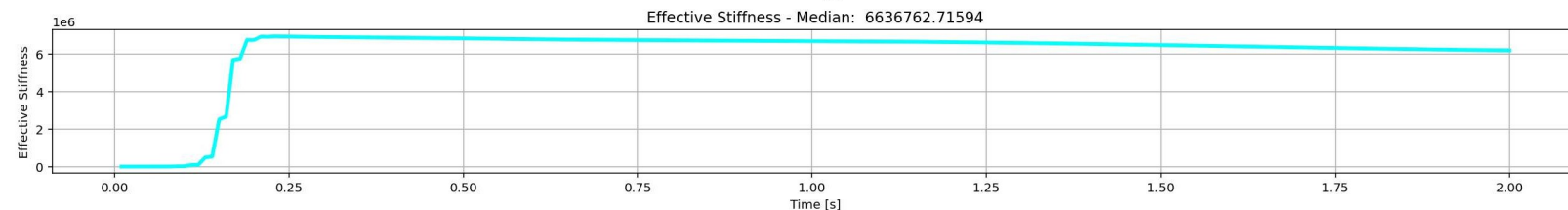
[Python Console] Plots Variable Explorer Help Debugger History Files



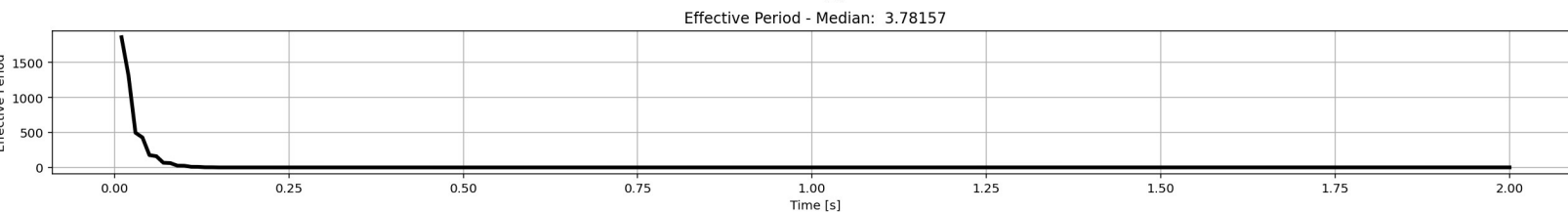
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



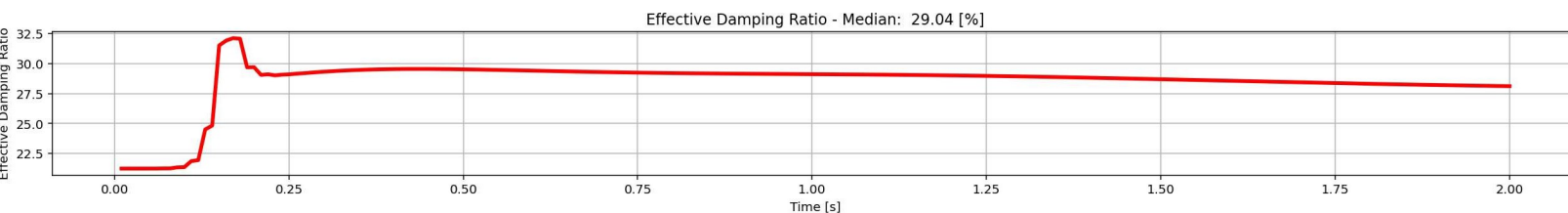
$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$



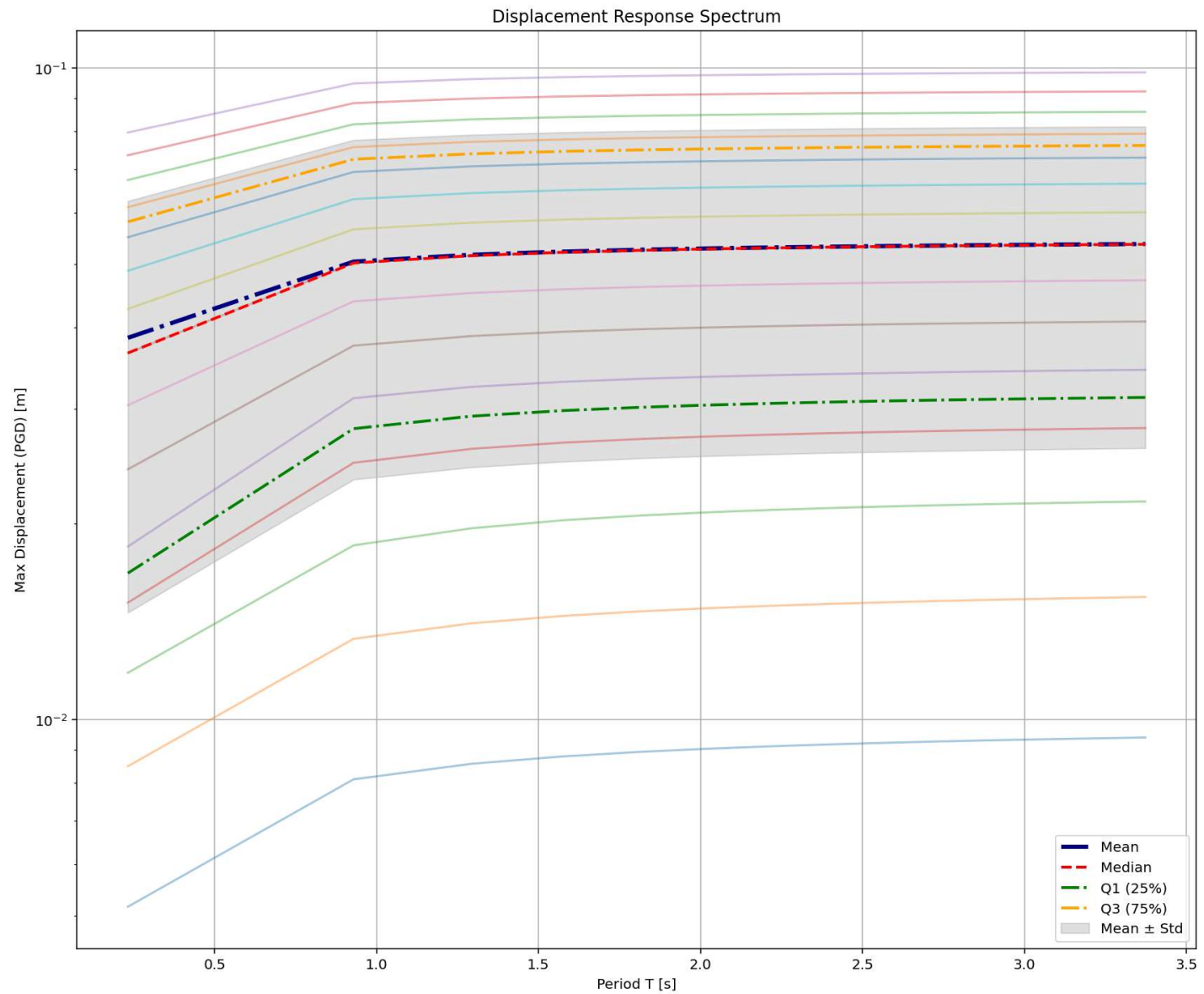
$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$

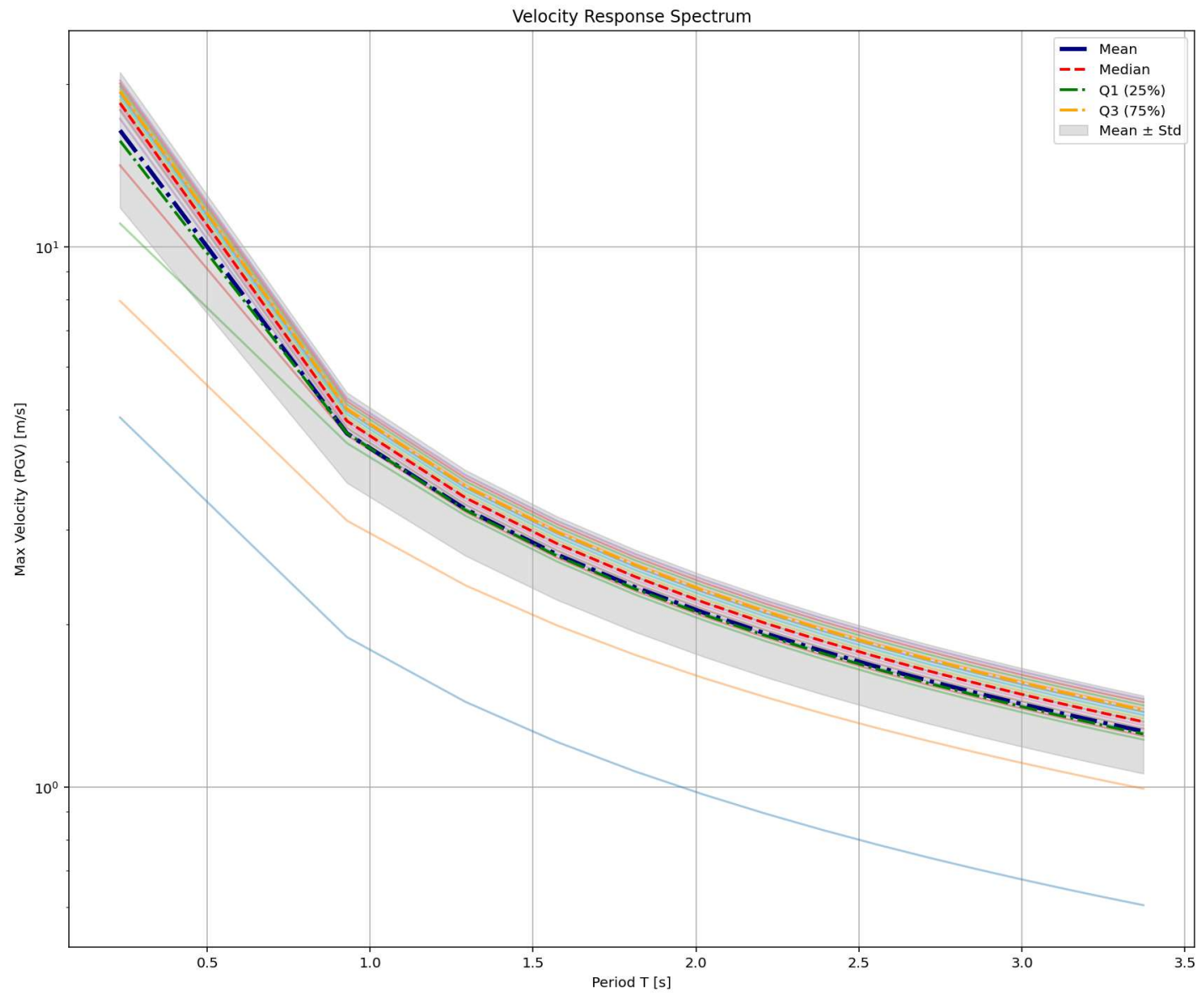


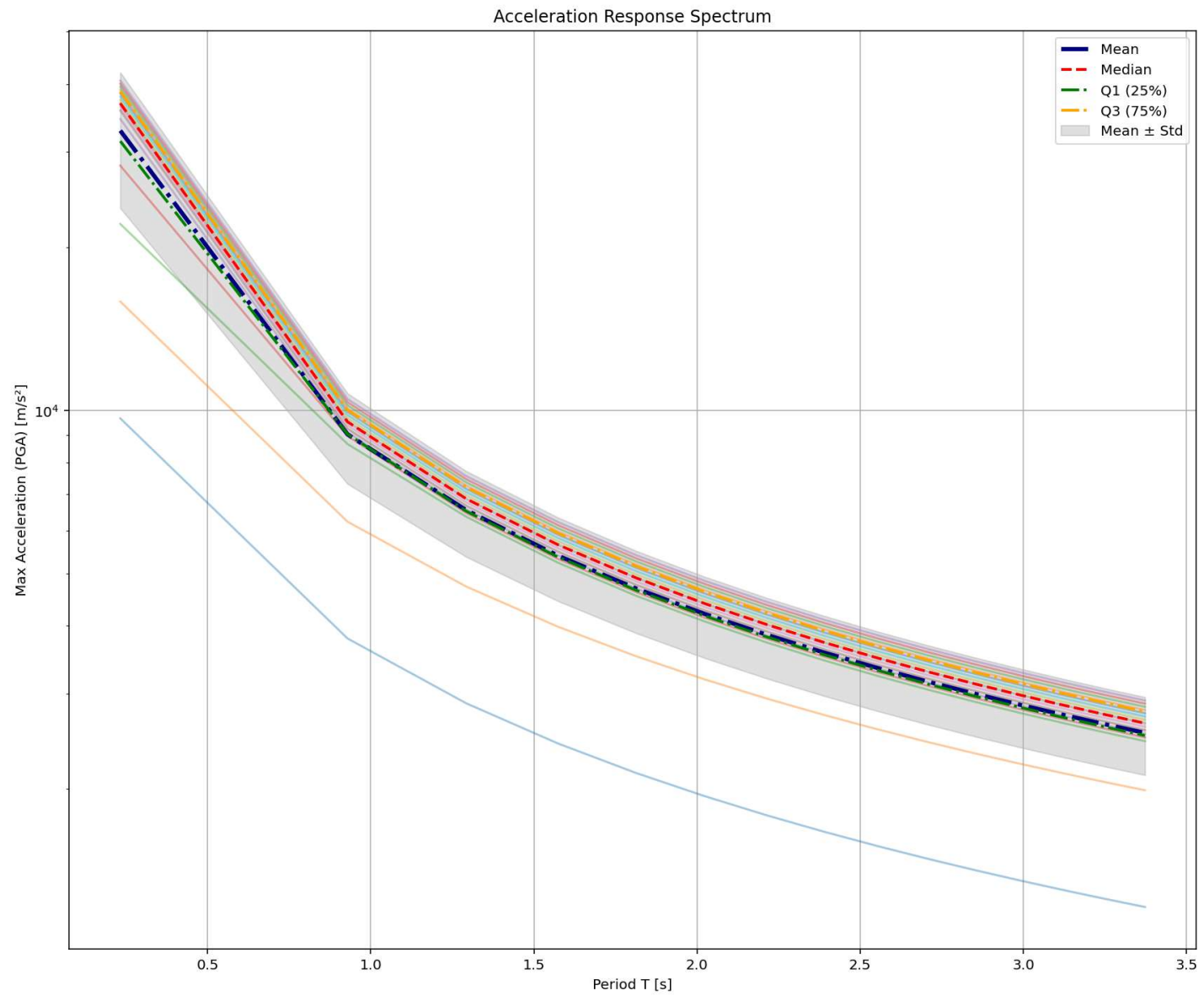
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

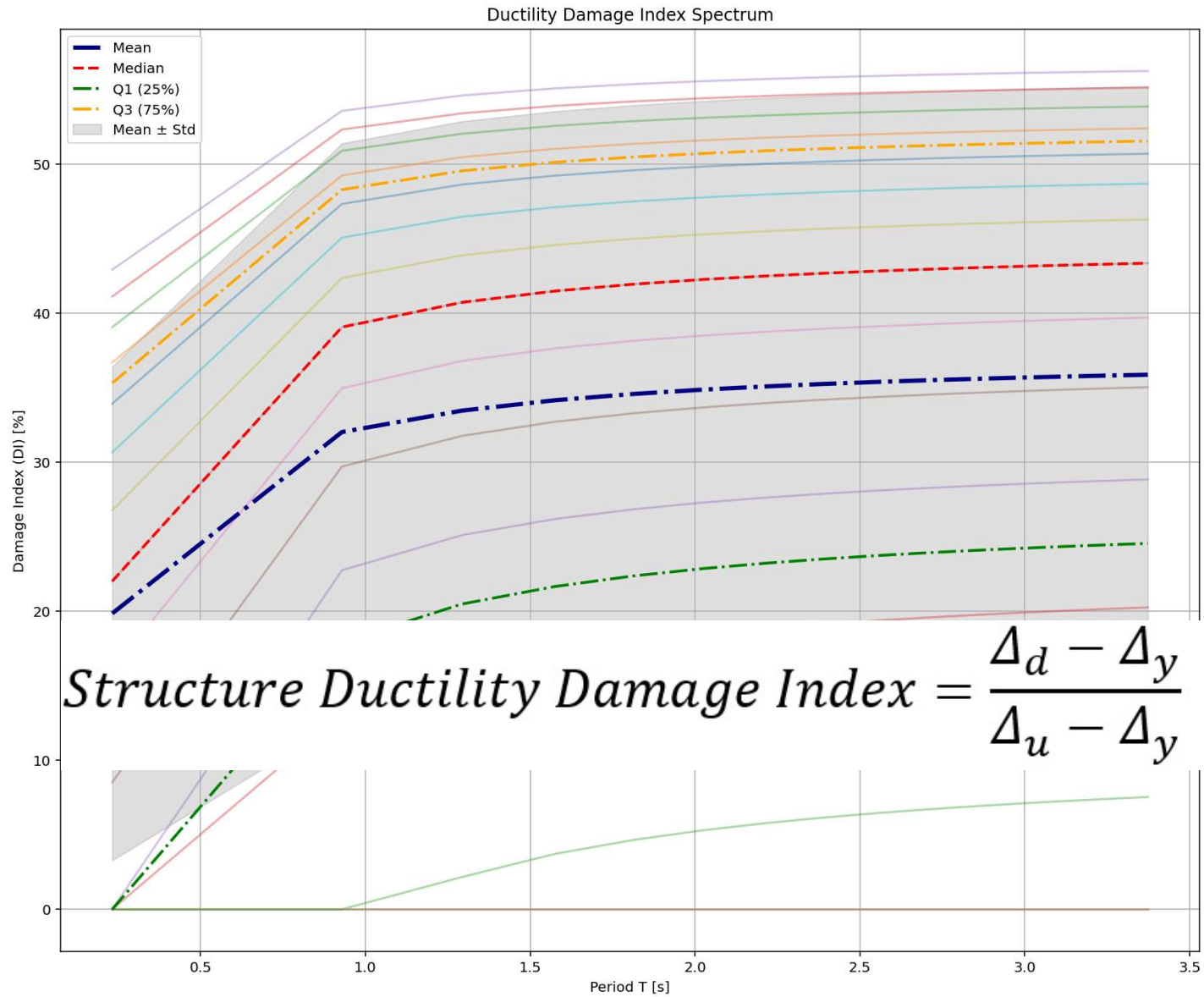


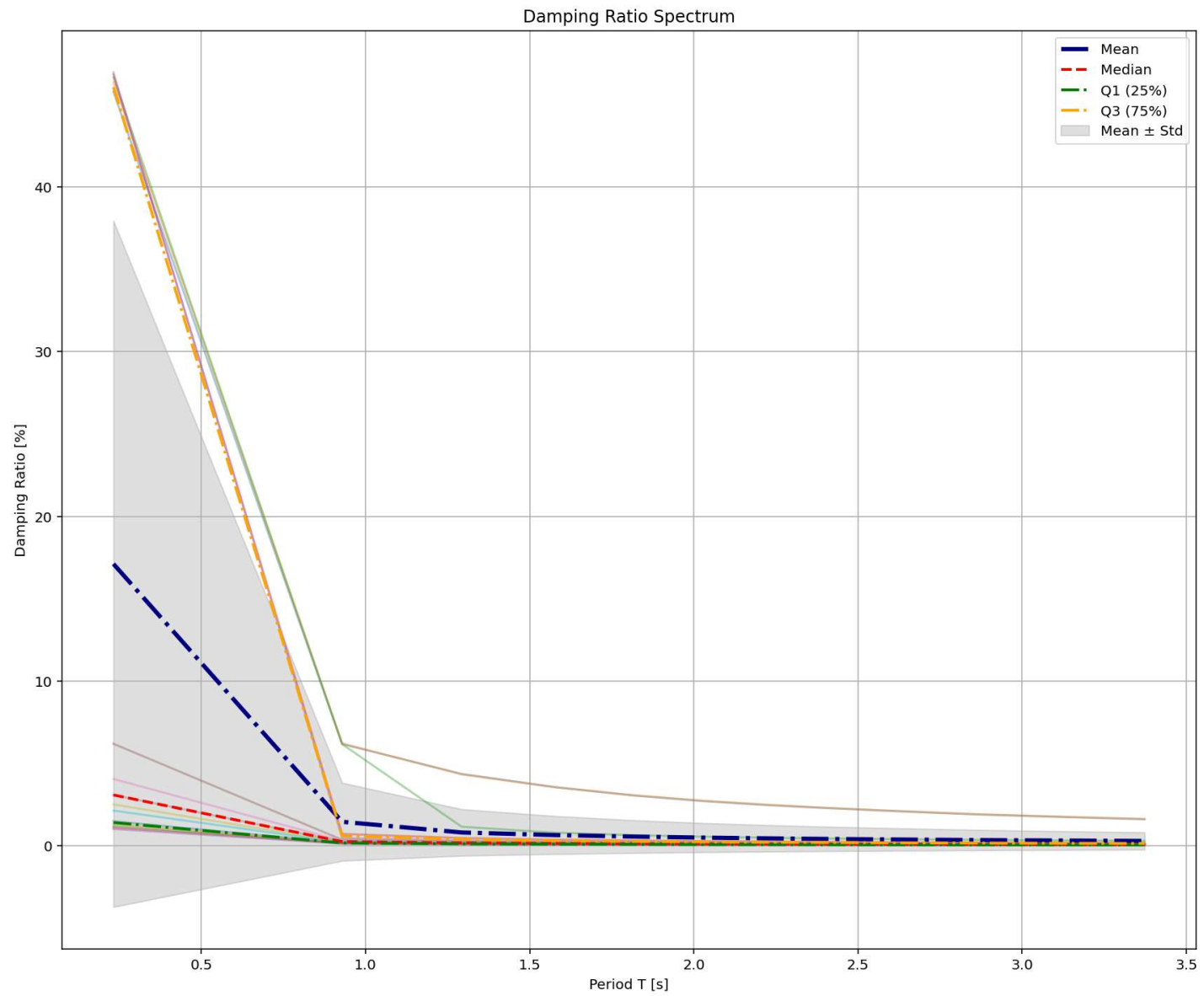
$$\xi_{eff} = \frac{\sum_{i=1}^n \xi_i \Delta_i}{\Delta_d}$$

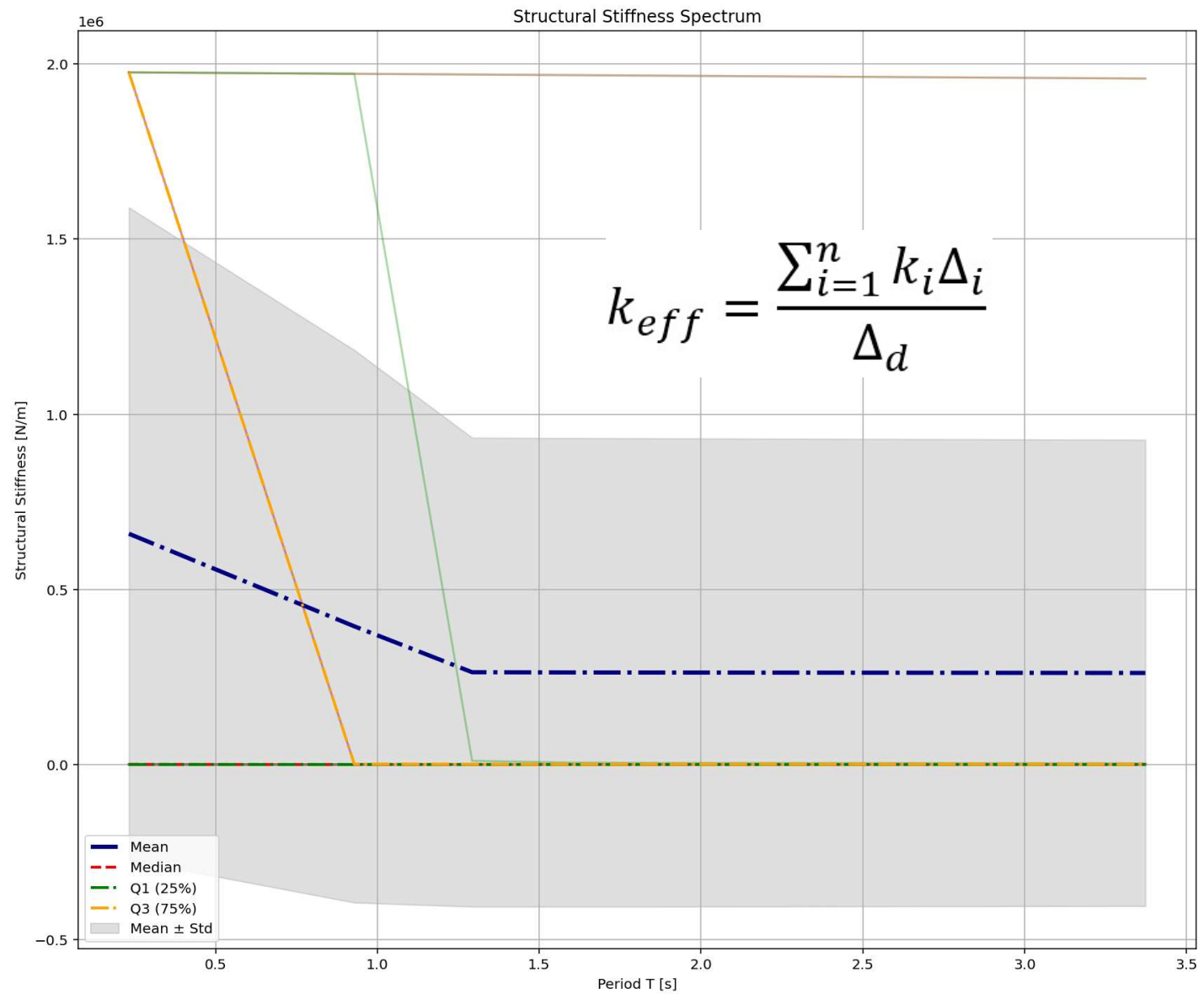


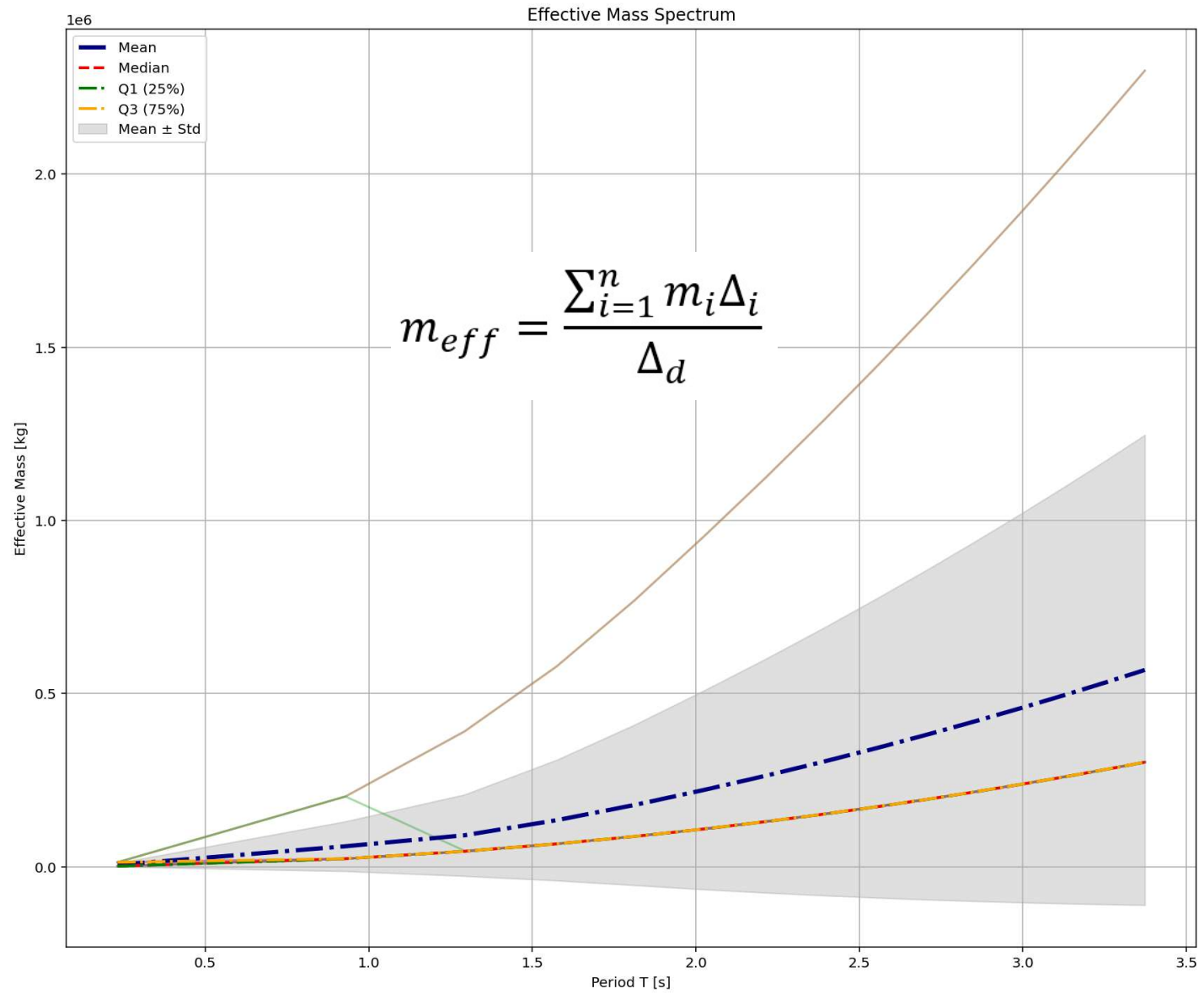


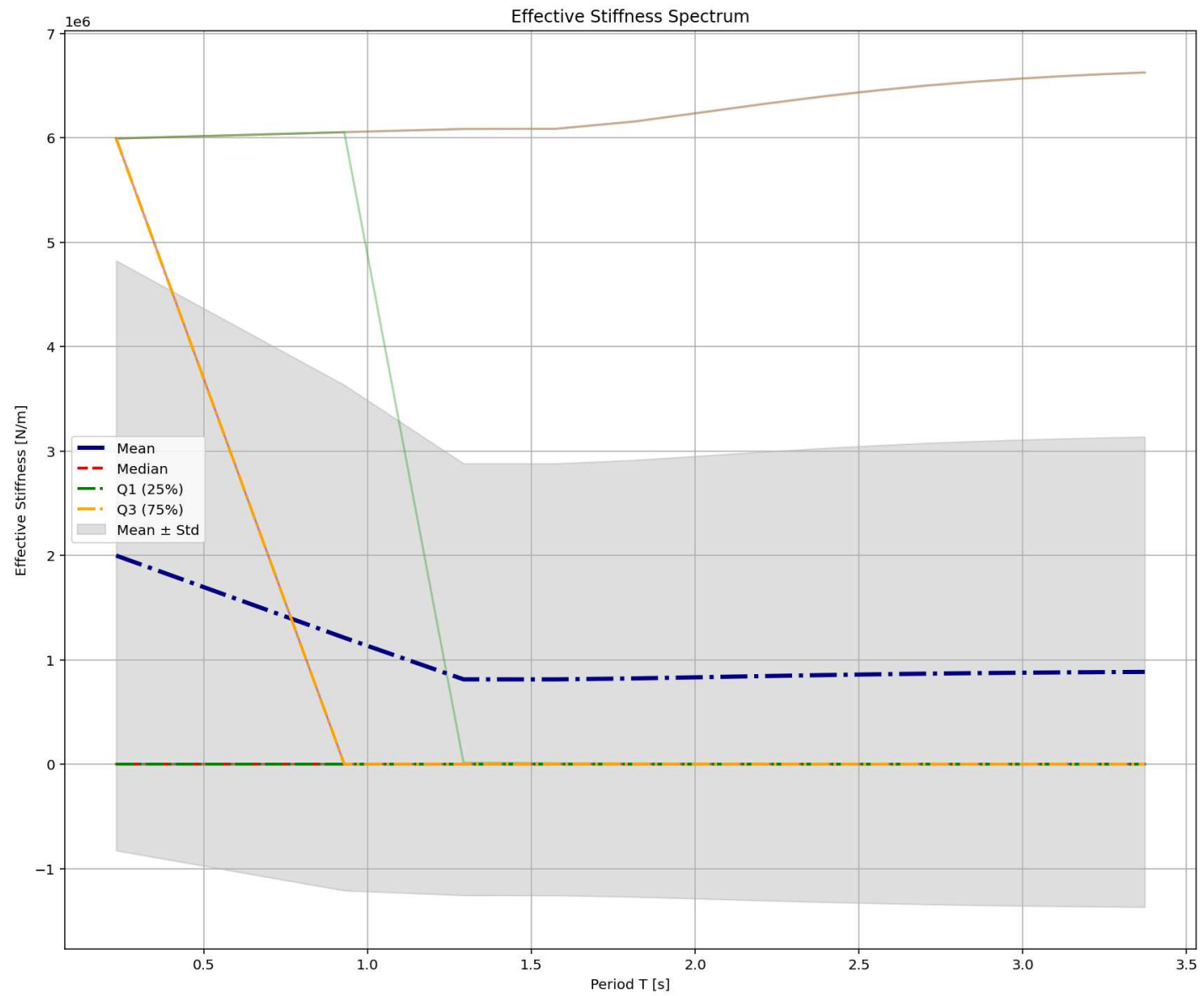


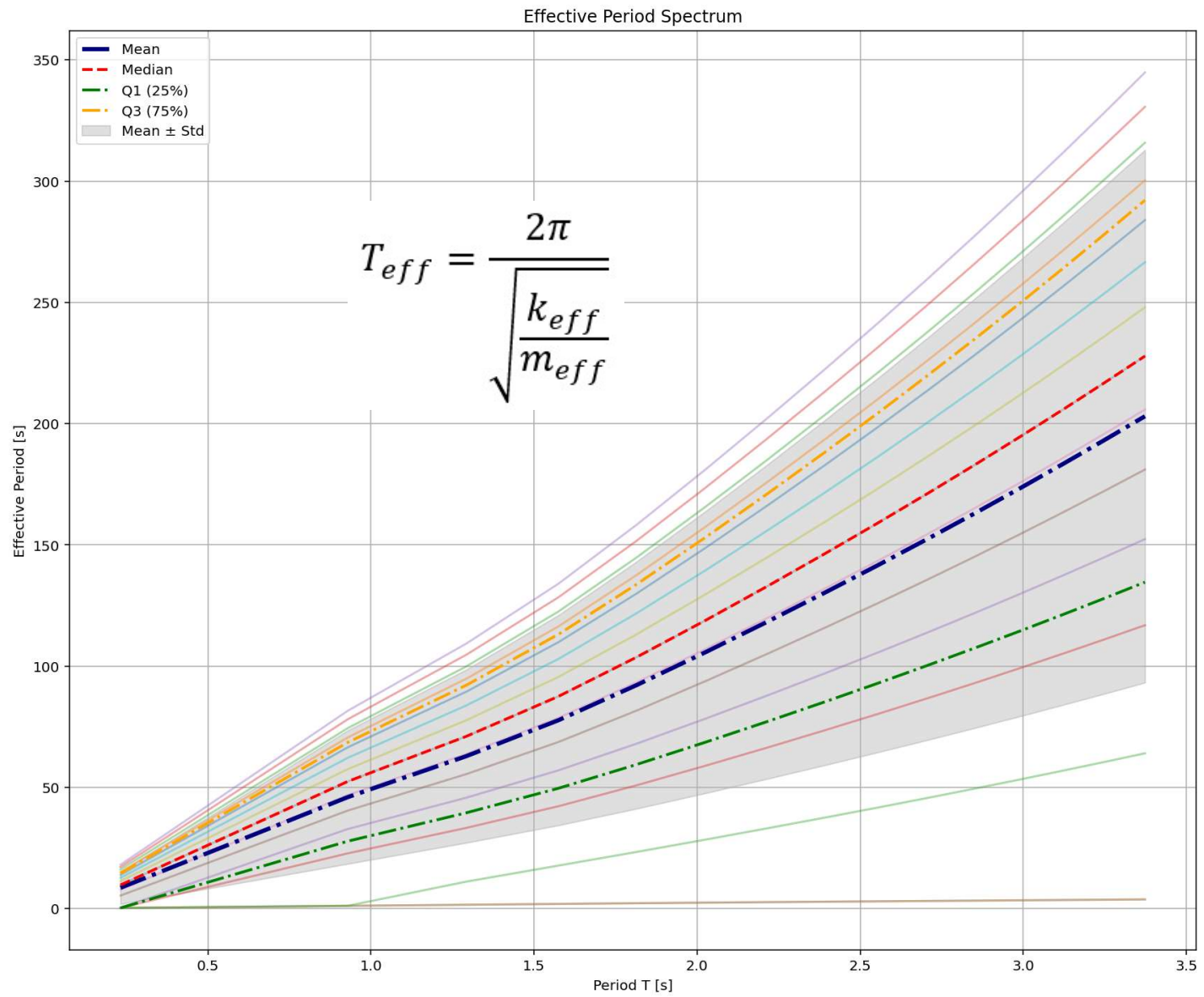


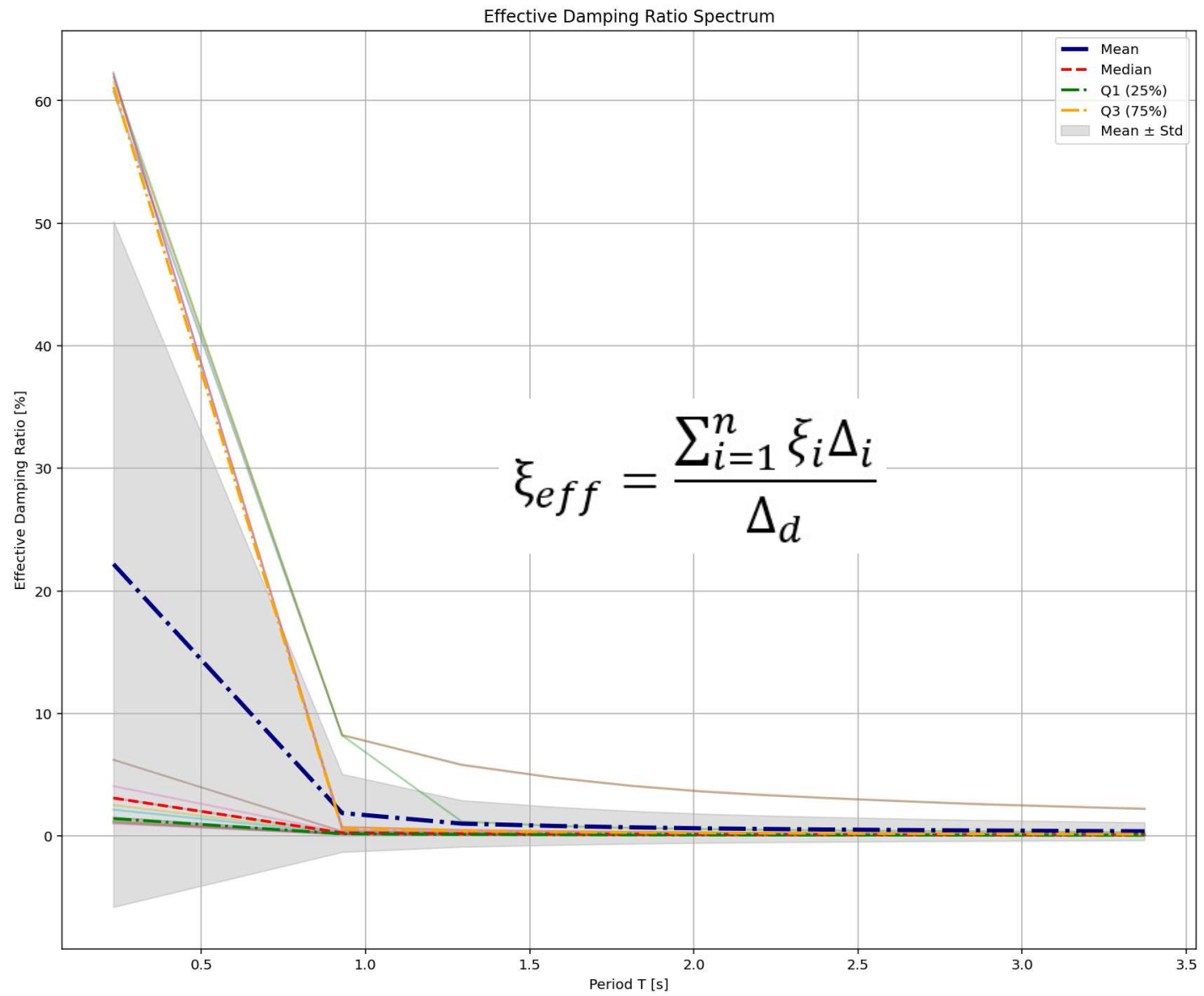


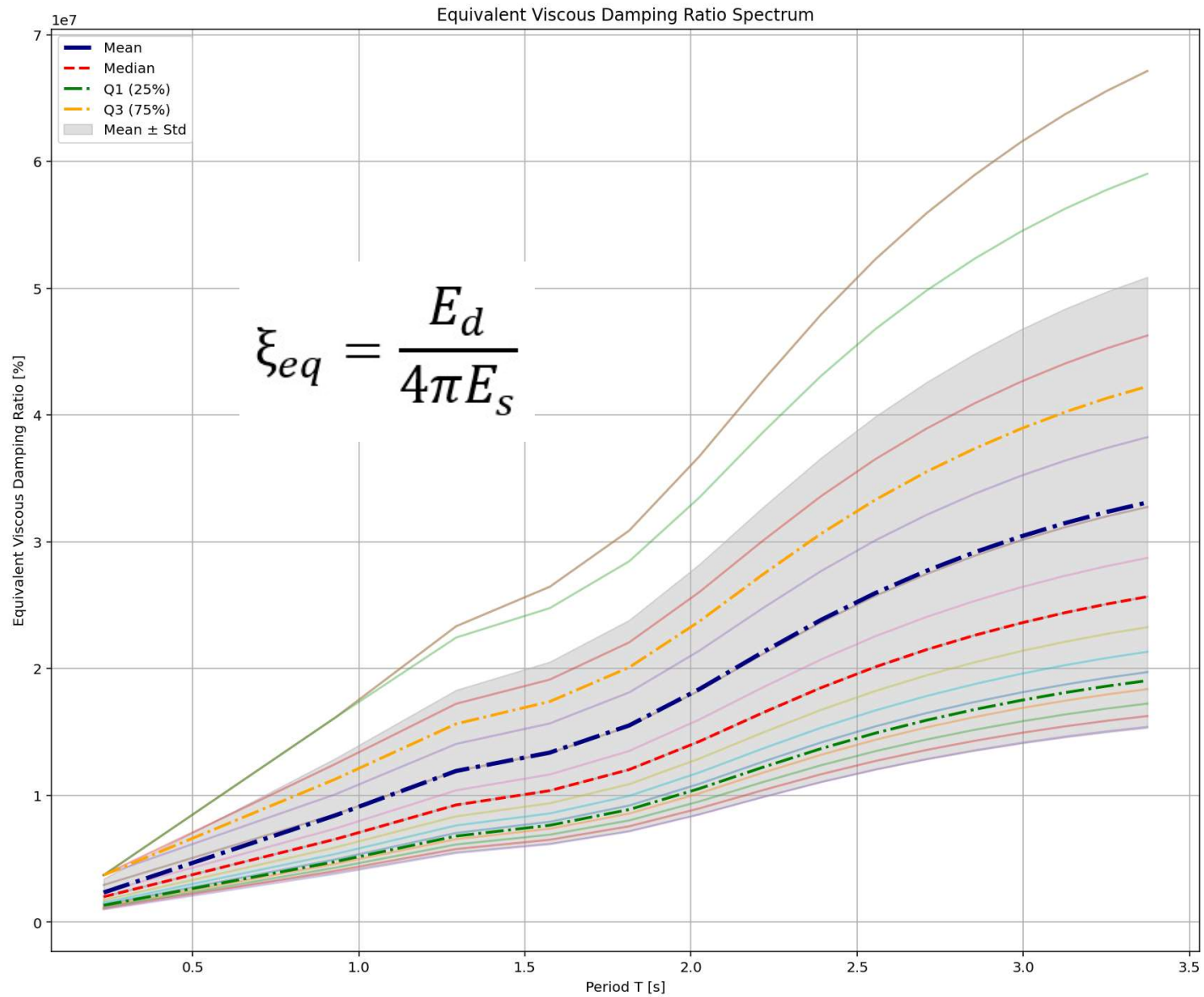


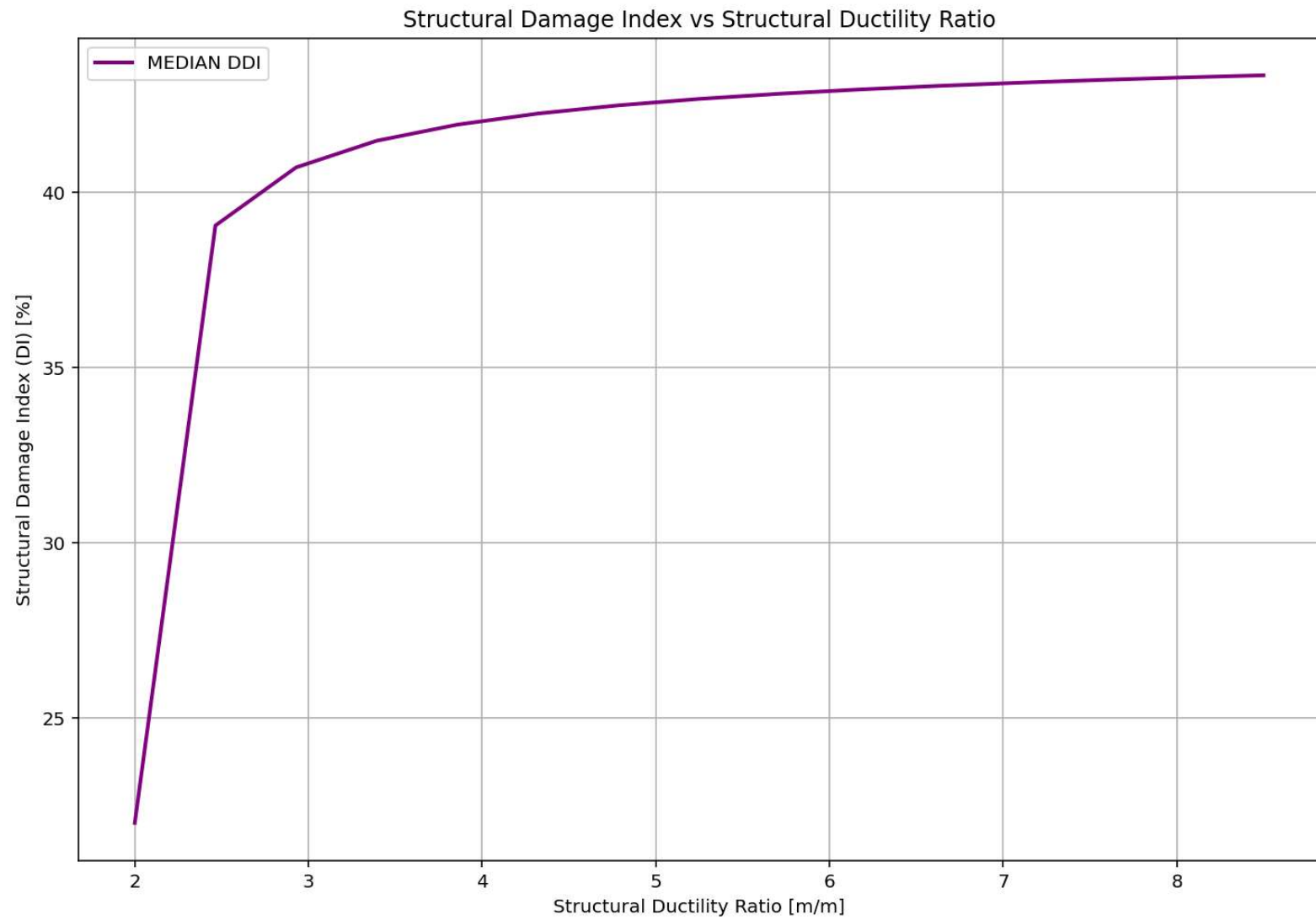


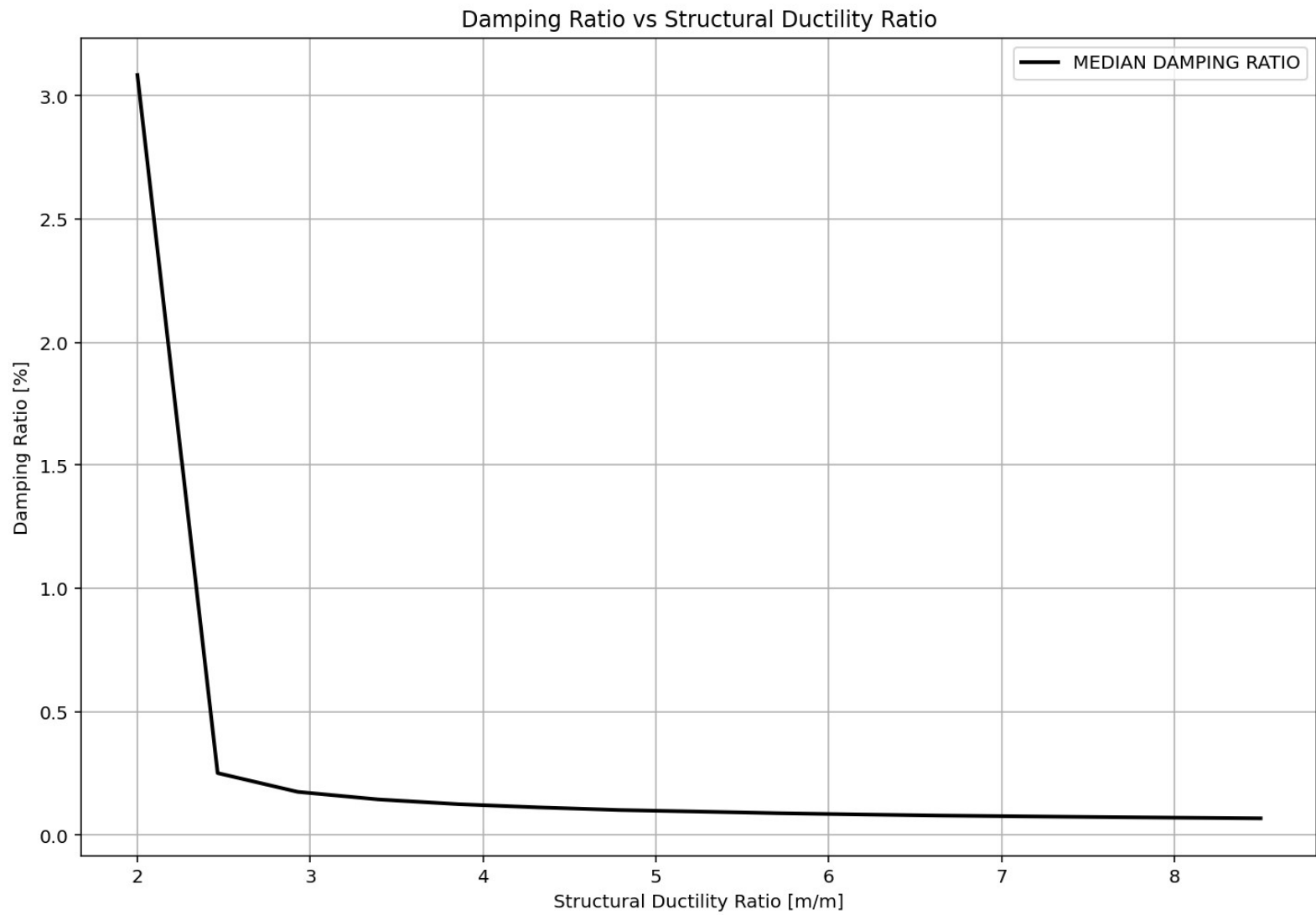


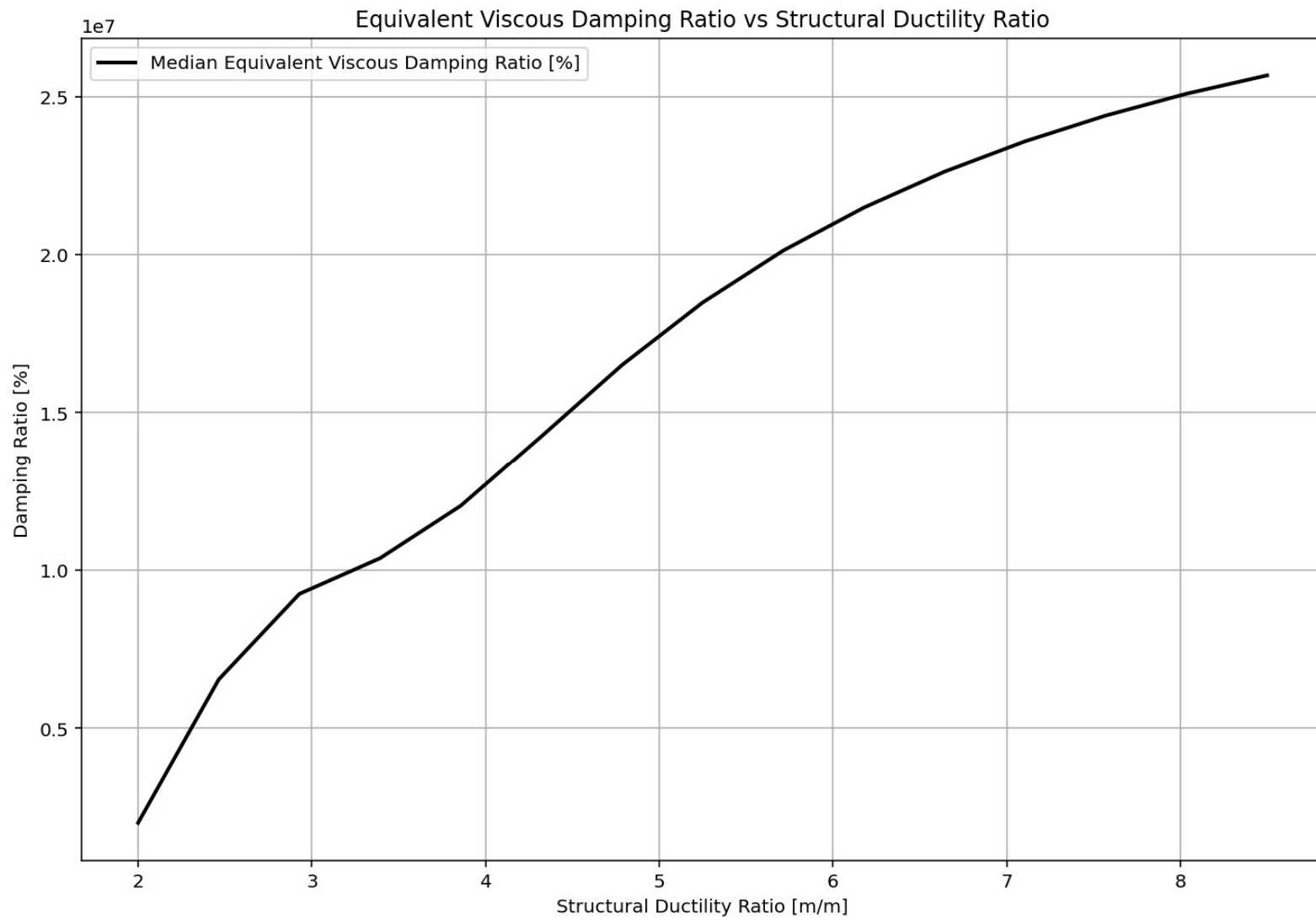




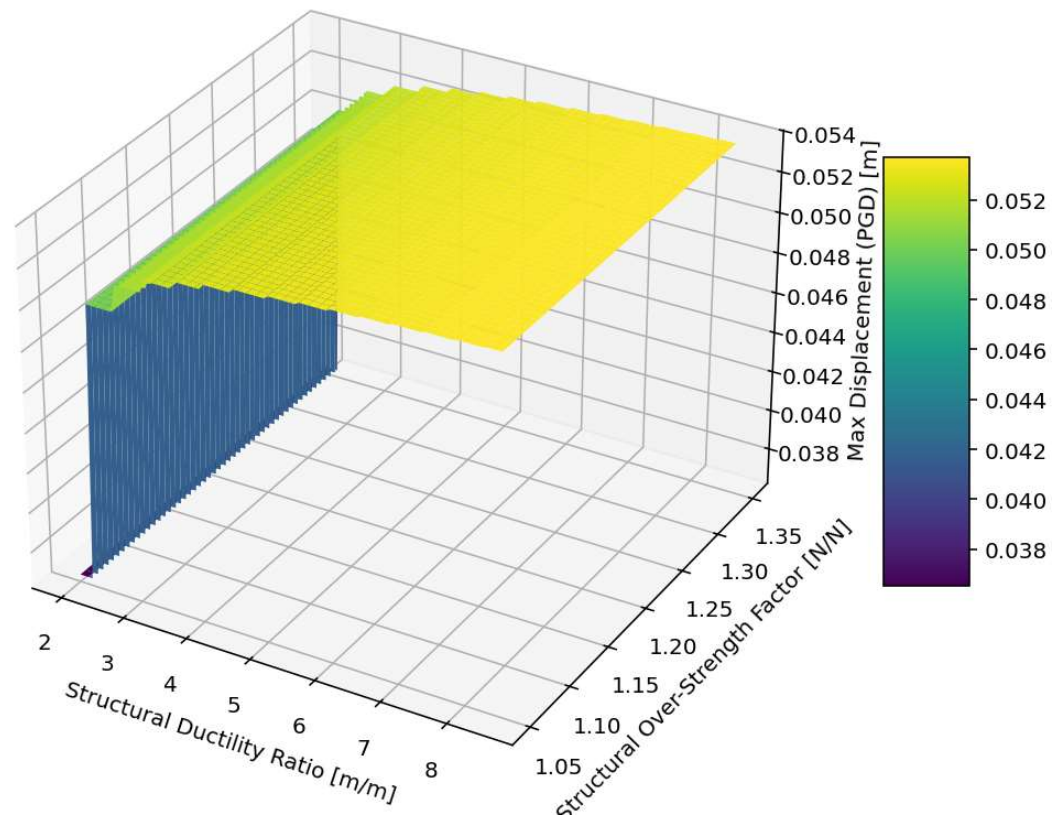




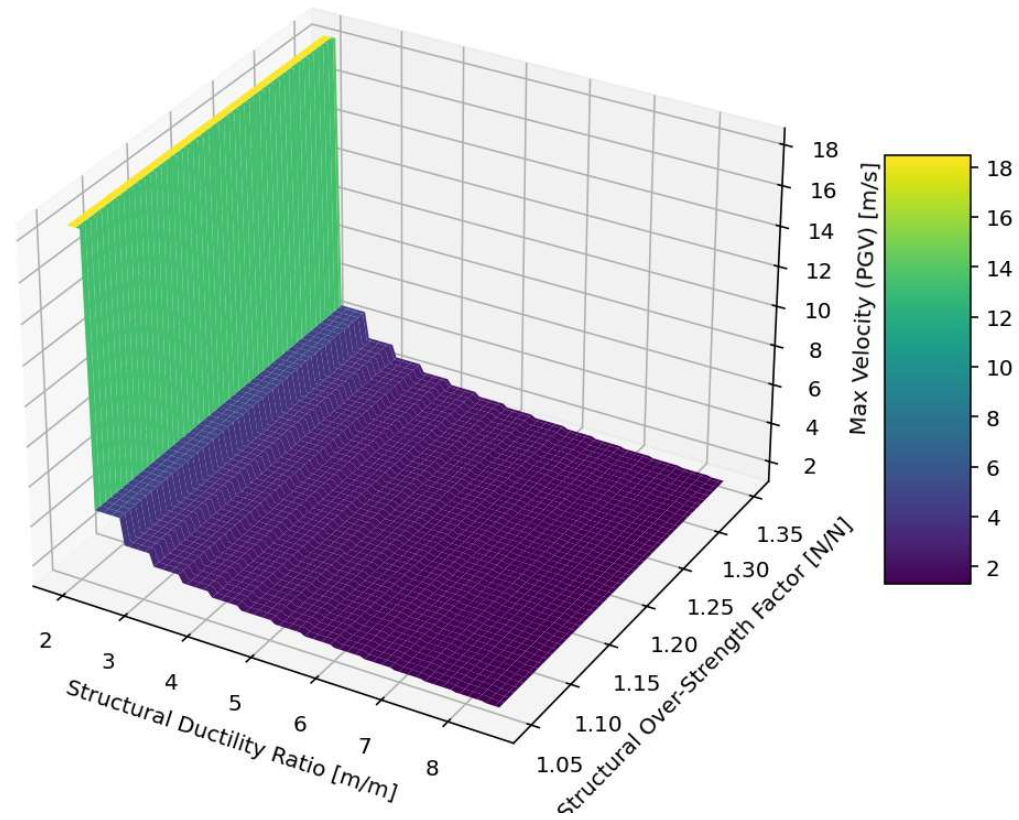




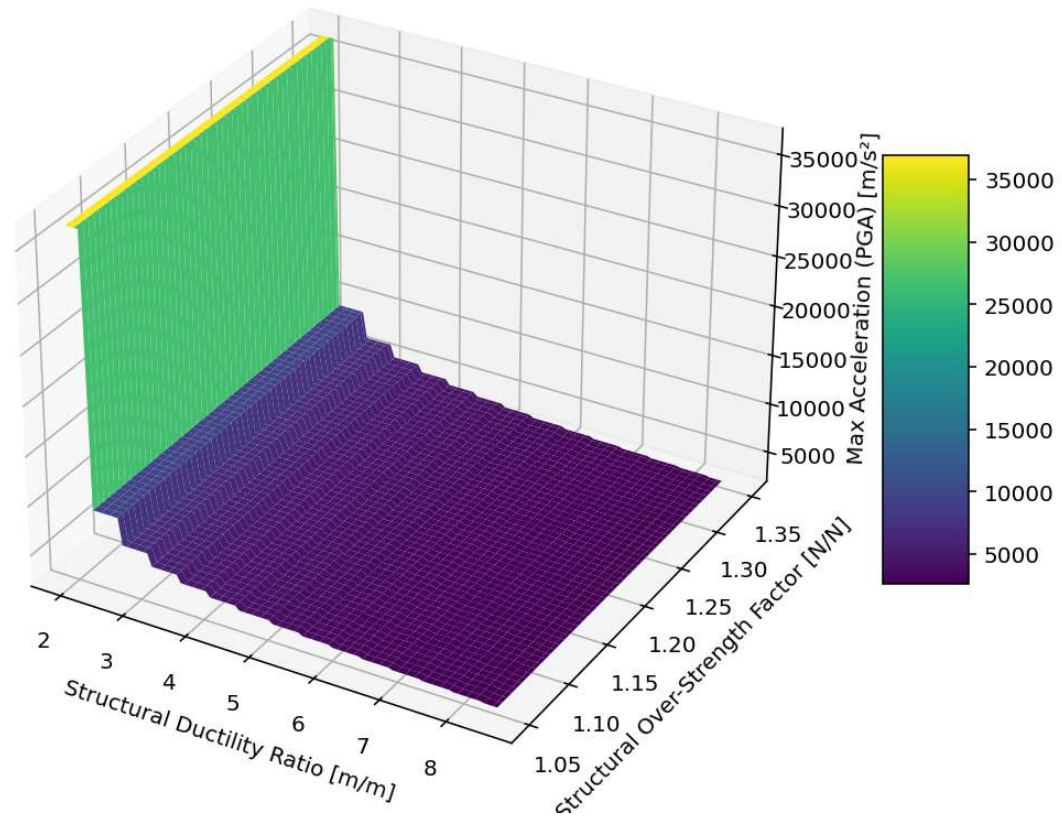
3D Contour Plot of Max Displacement (PGD) [m]



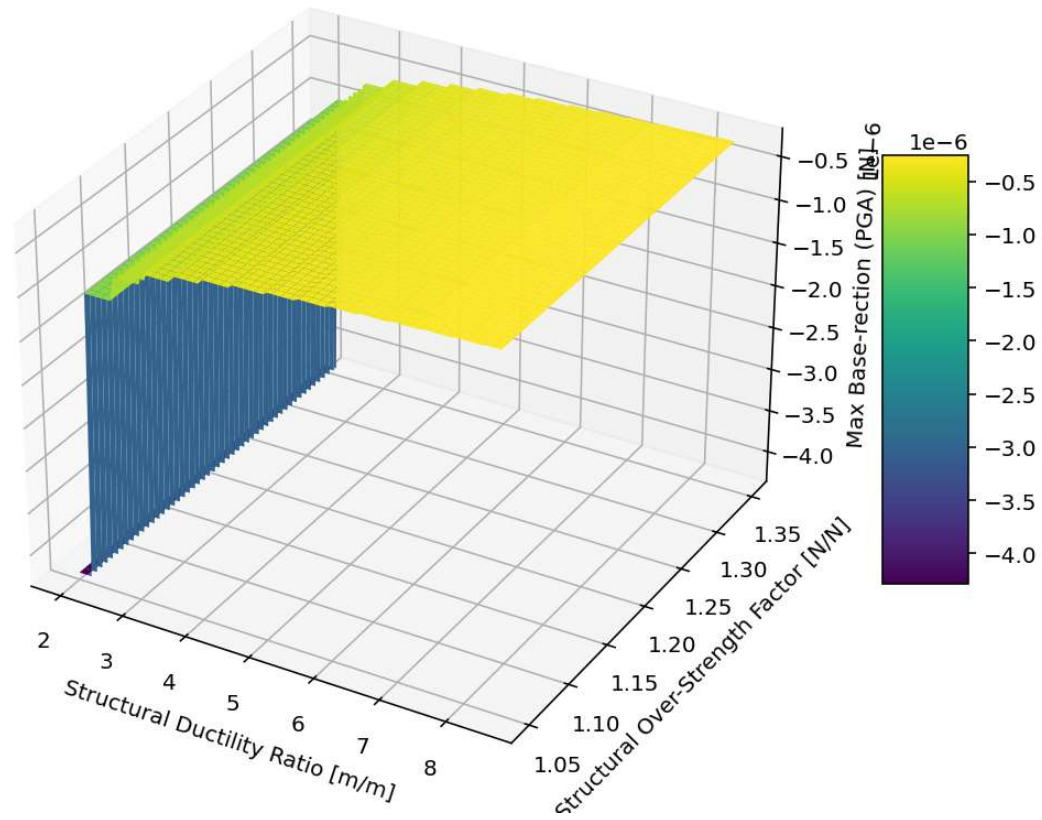
3D Contour Plot of Max Velocity (PGV) [m/s]



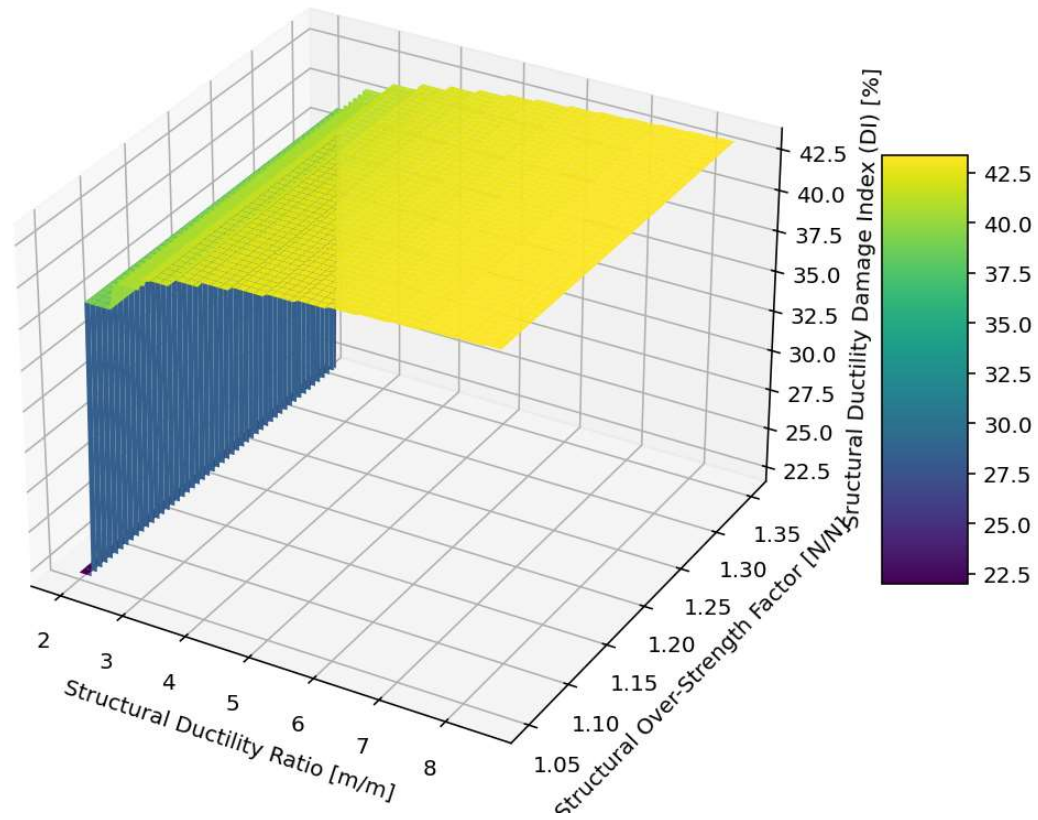
3D Contour Plot of Max Acceleration (PGA) [m/s²]



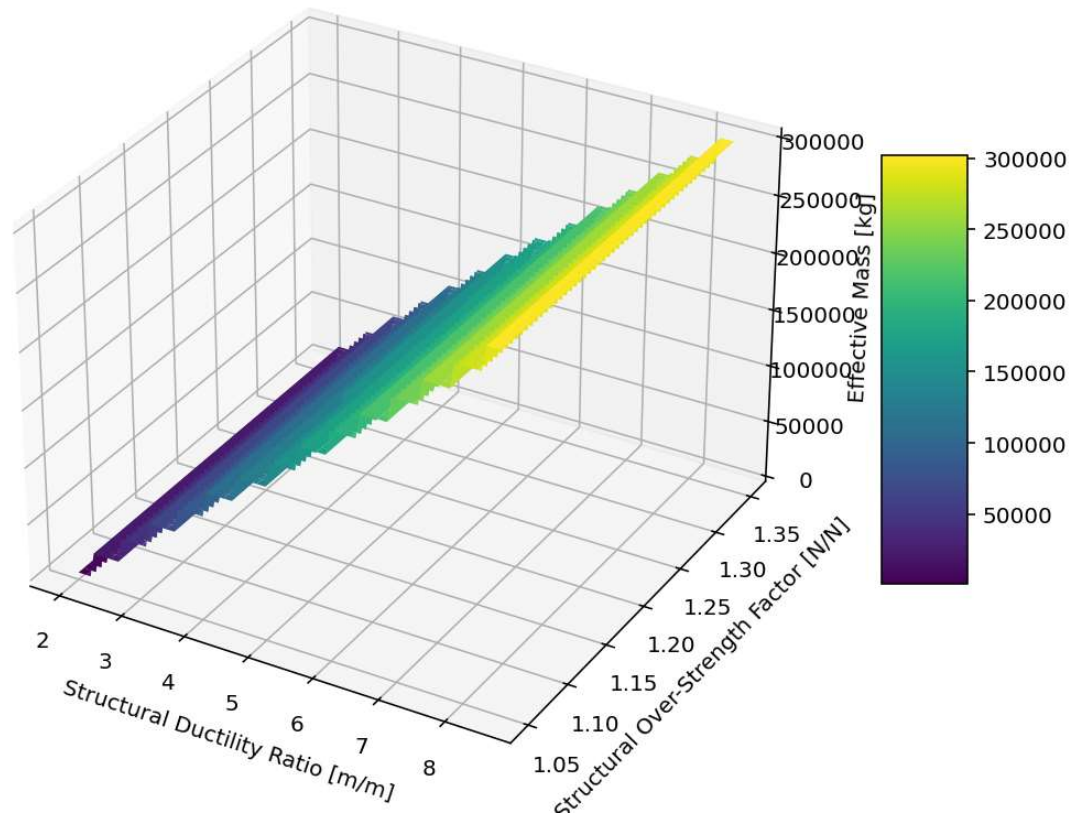
3D Contour Plot of Max Base-rection (PGA) [N]



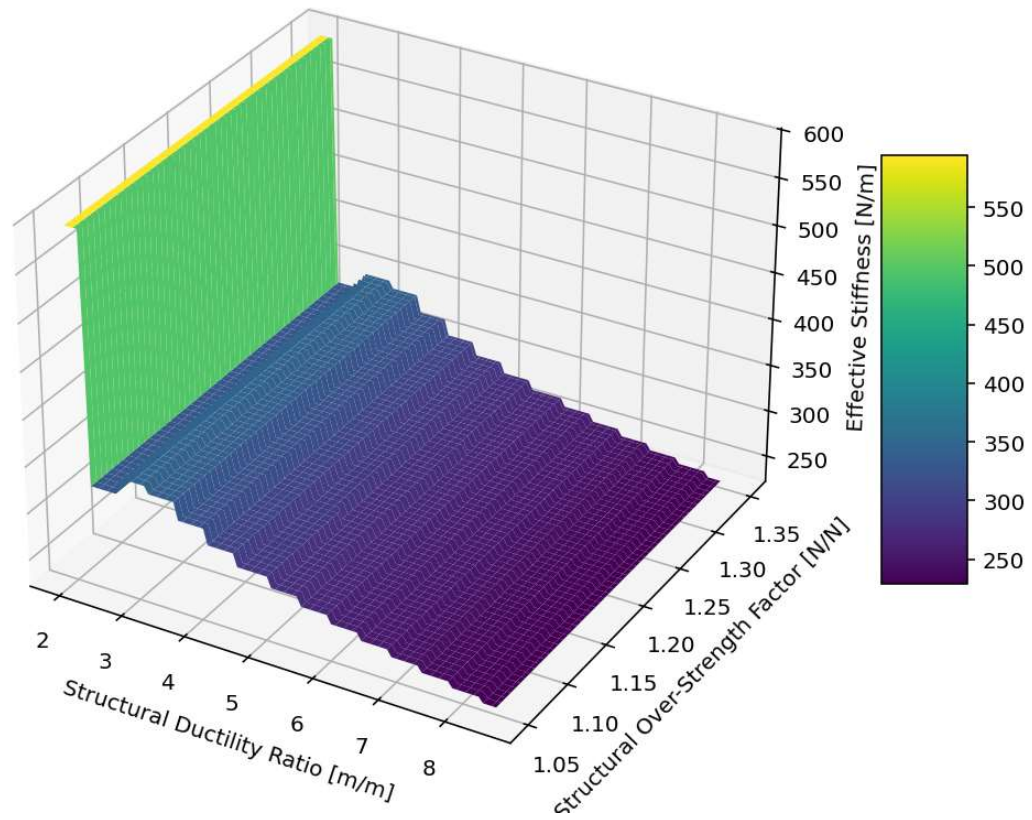
3D Contour Plot of Structural Ductility Damage Index (DI) [%]



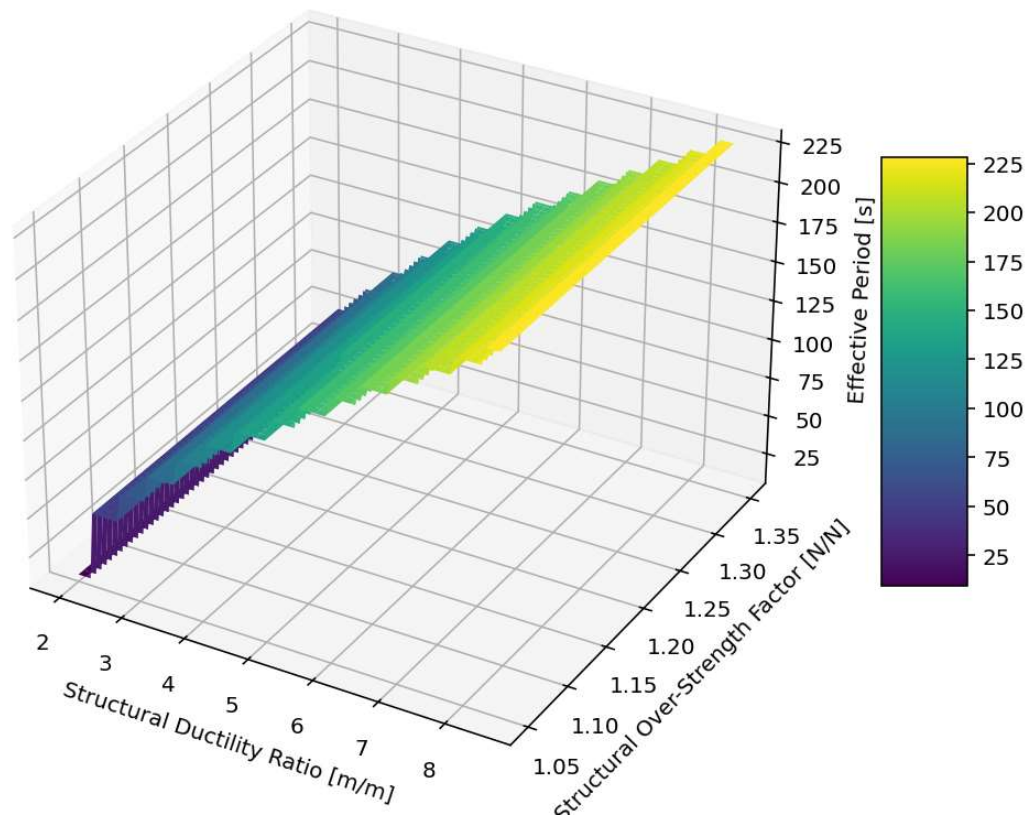
3D Contour Plot of Effective Mass [kg]



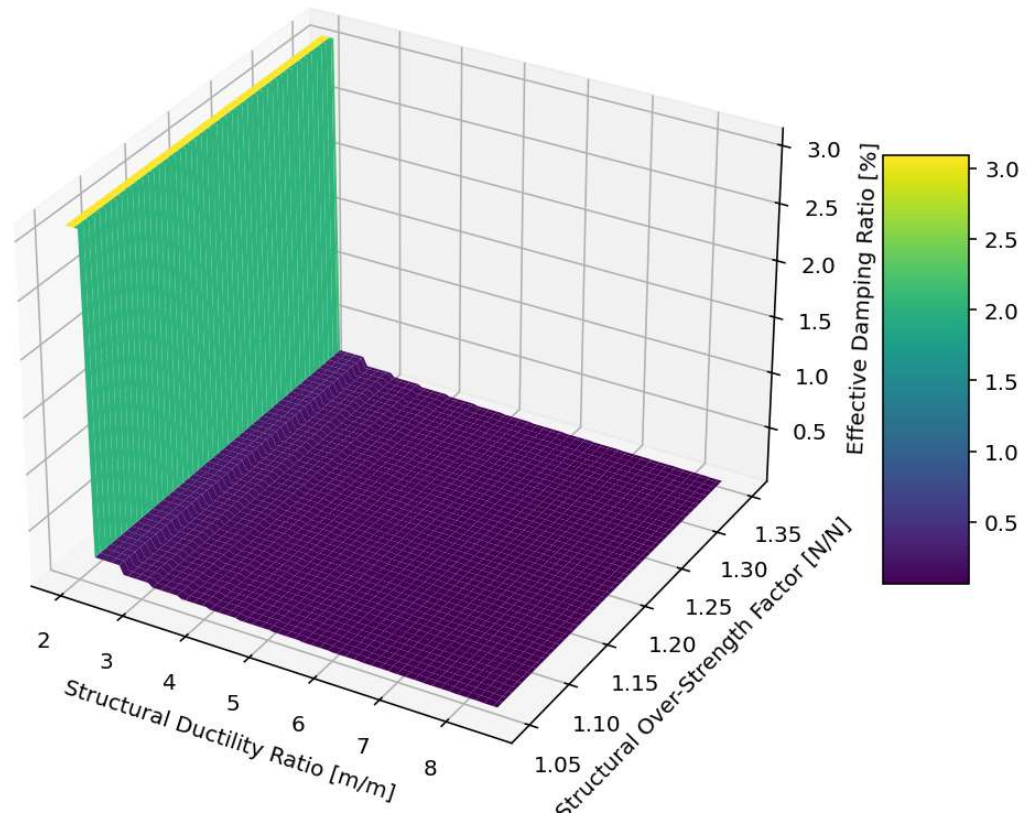
3D Contour Plot of Effective Stiffness [N/m]



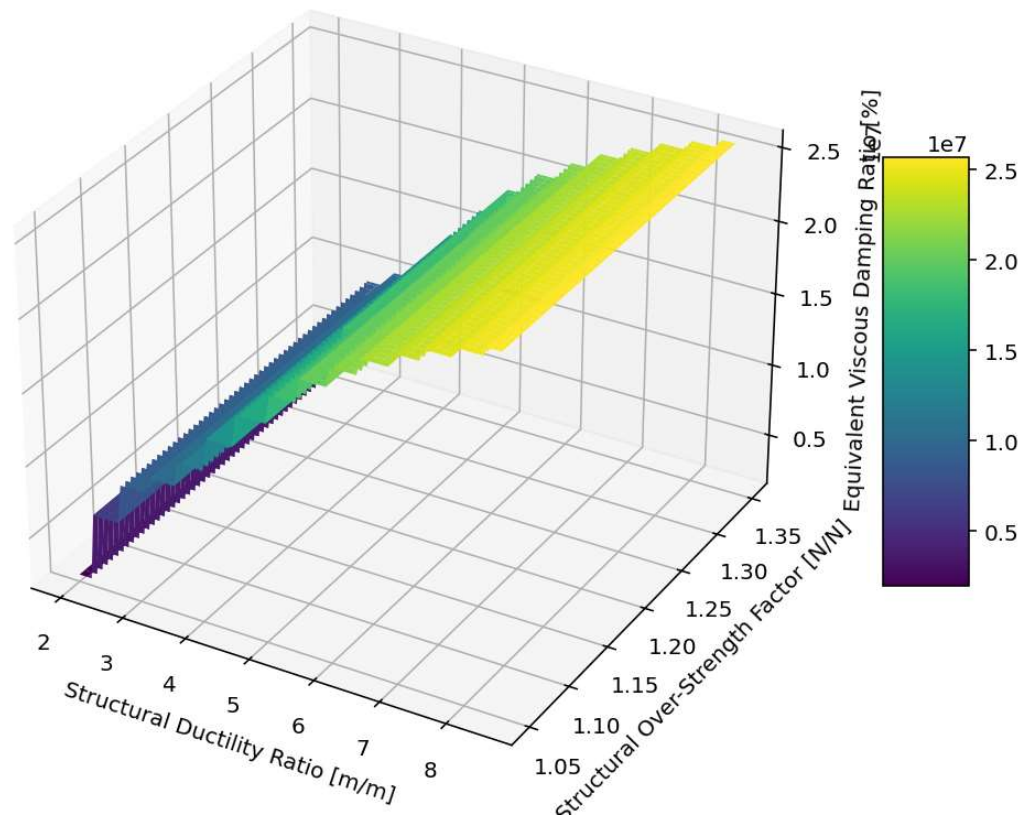
3D Contour Plot of Effective Period [s]



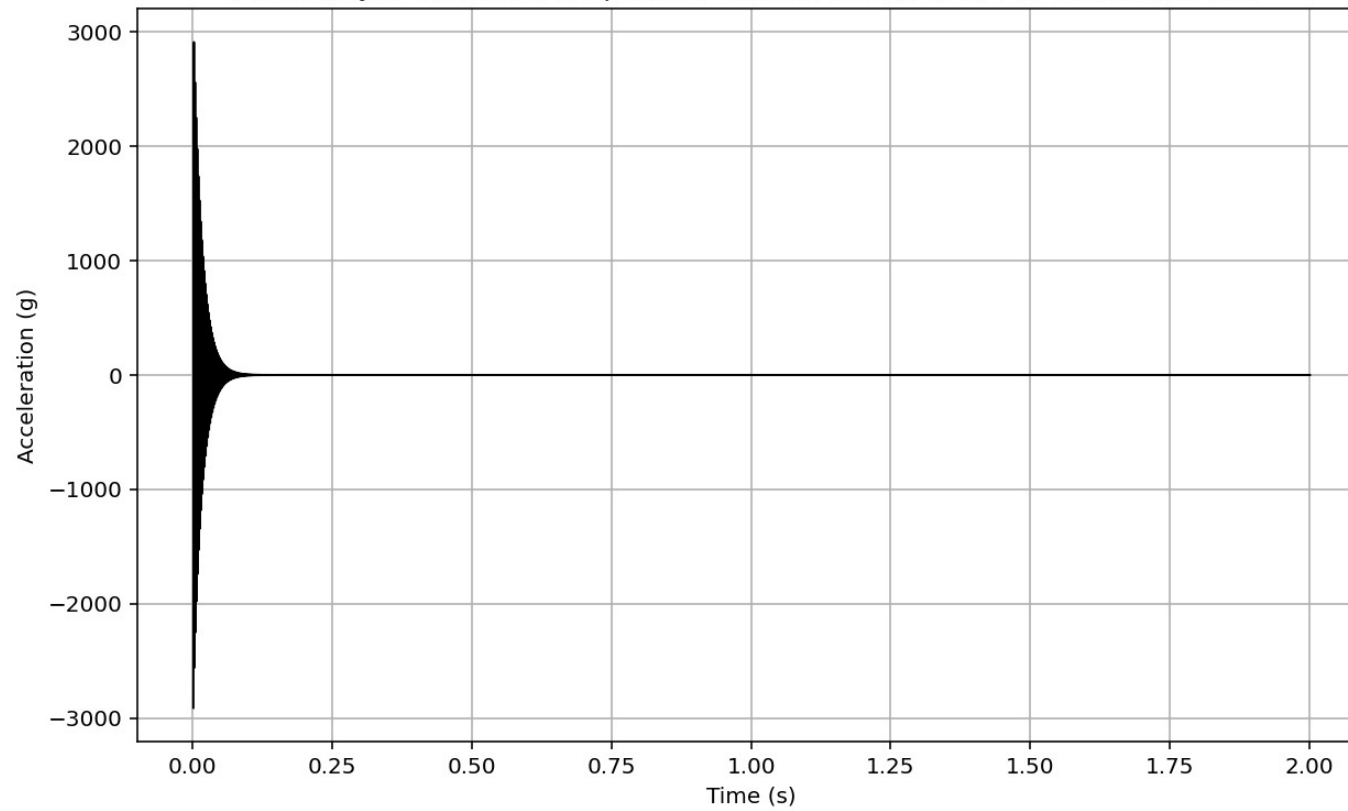
3D Contour Plot of Effective Damping Ratio [%]

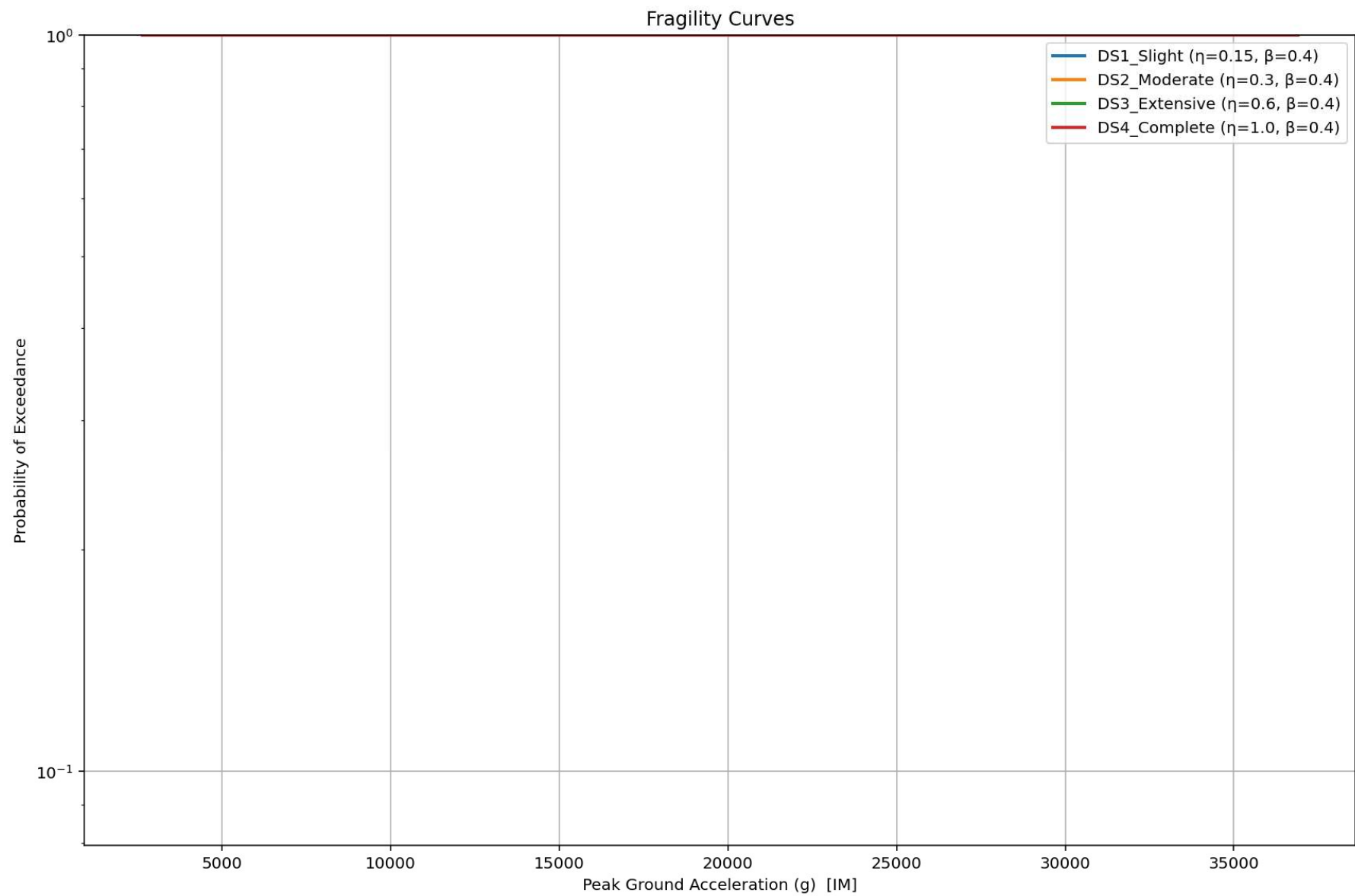


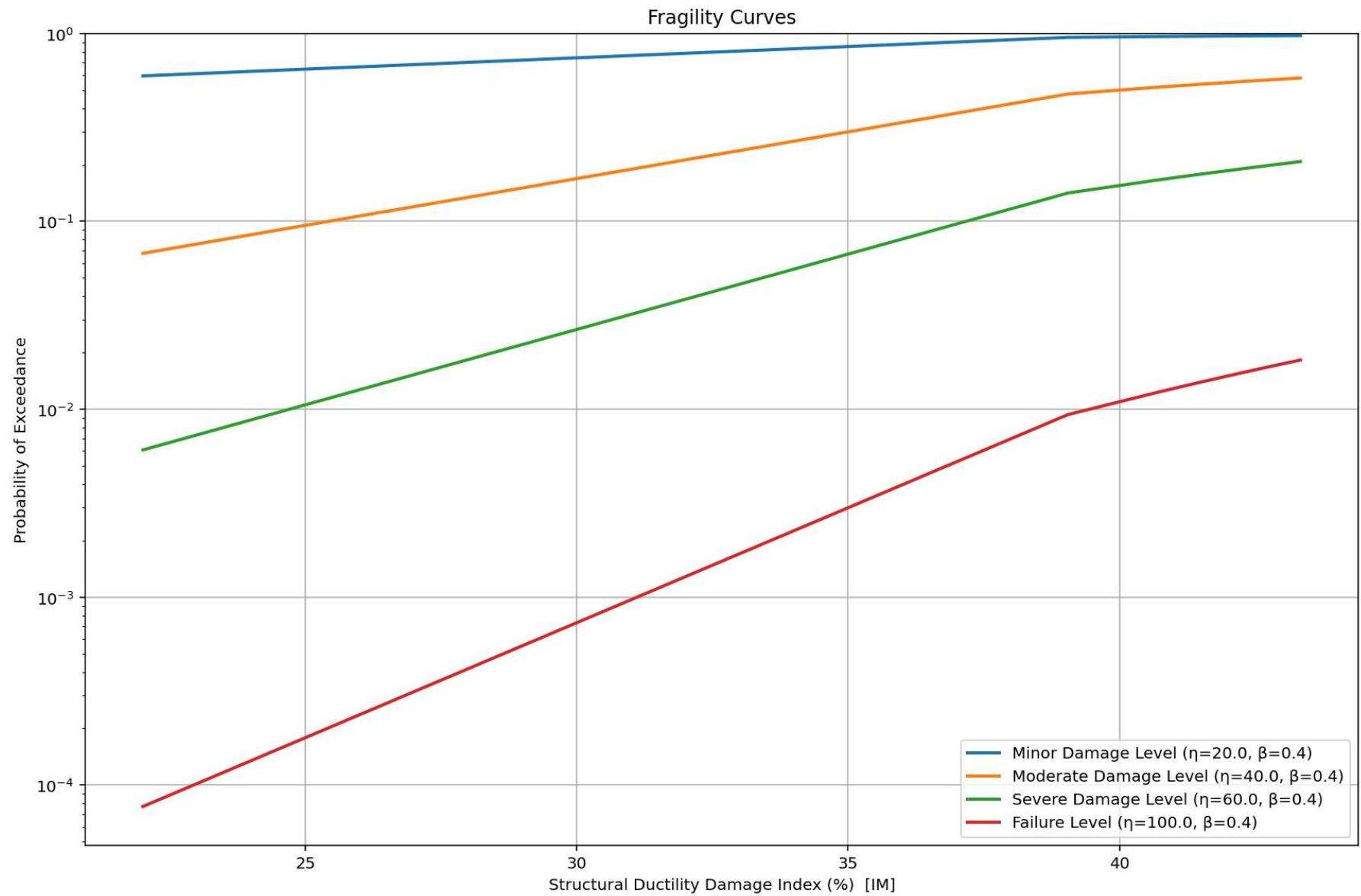
3D Contour Plot of Equivalent Viscous Damping Ratio [%]

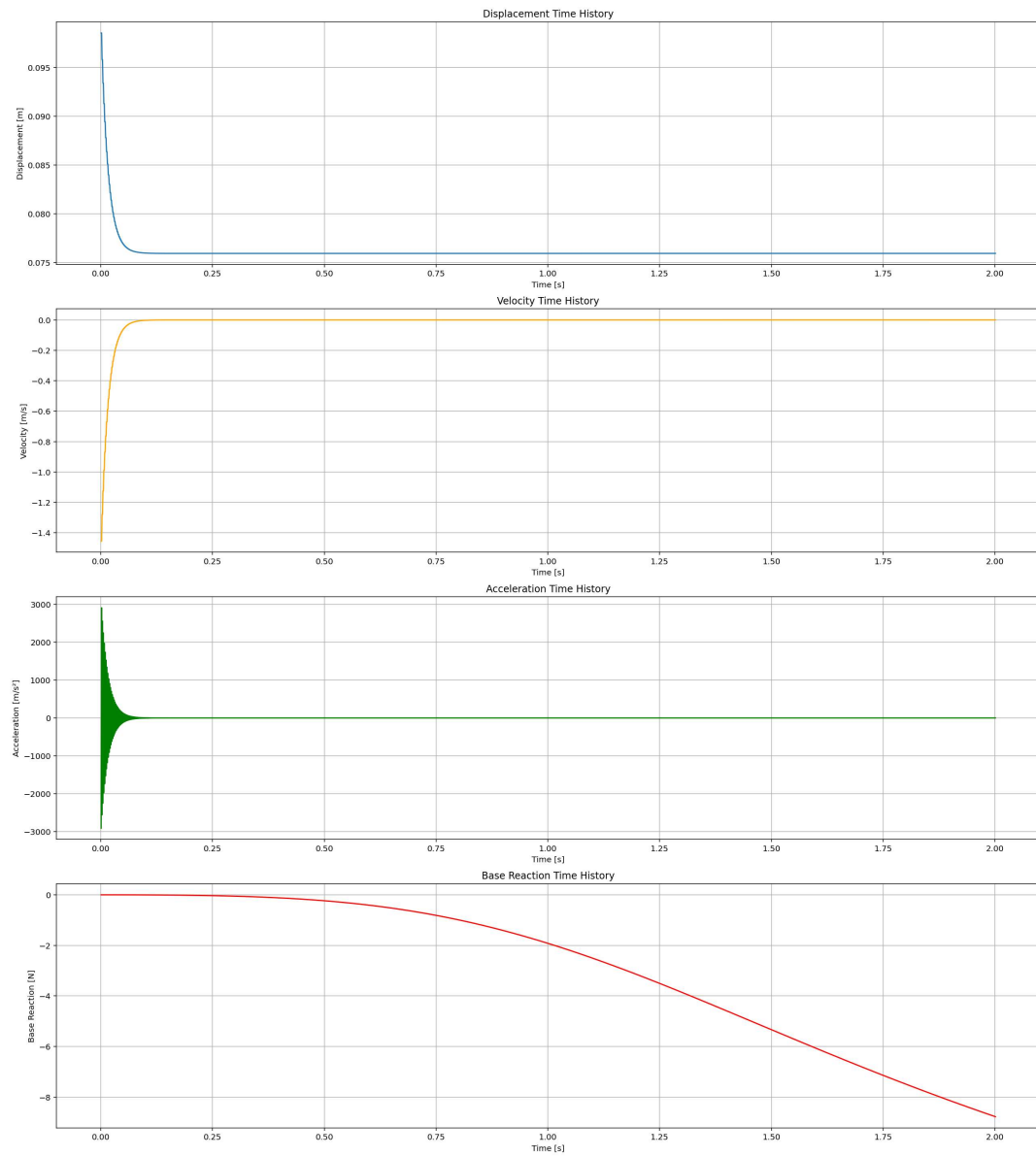


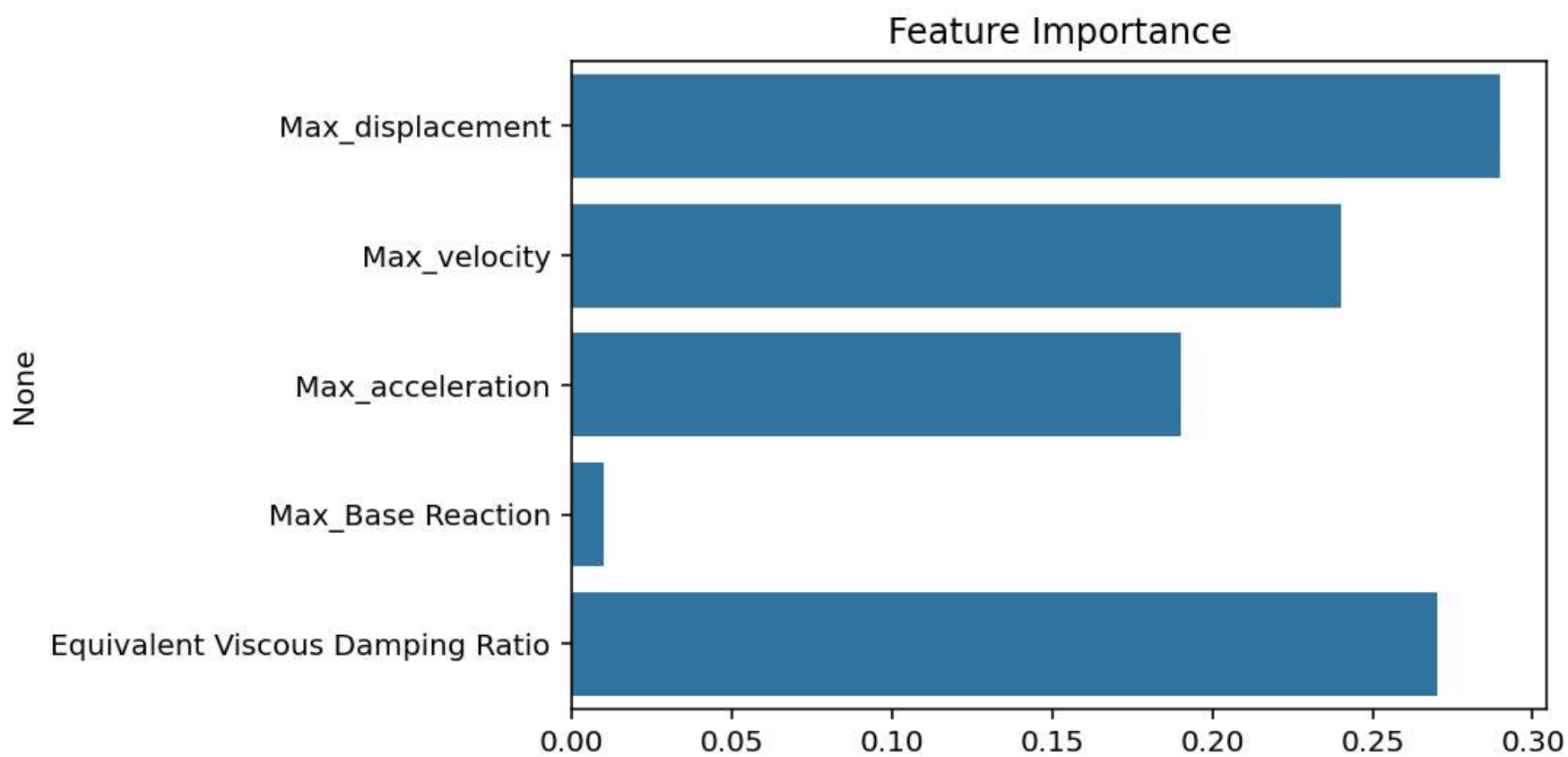
Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 2911.4193











Correlation Heatmap

